Notes of Optimization

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Contents

Chapter 1

Lecture 1 of 03/03/2026

We will give a quick summary on basic notions of topology, calculus 1 & 2.

Definition 1 (Norm/scalar product)**.** Given $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ the Euclidian norm (or ℓ^2 norm) is defined as

$$\|x\| = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2}.$$

For $x, y \in \mathbb{R}^n$ the *scalar product* is given by

$$\langle x, y \rangle = x \cdot y = x^\top y = \sum_{i=1}^{n} x_i y_i.$$

Moreover, the scalar product induces a norm. Indeed, one can verify that

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Definition 2 (Open ball)**.** Given $x \in \mathbb{R}^n$, $r > 0$ we call the *open ball of radius r centered in* x the set

$$B(x, r) = \{ y \in \mathbb{R}^n : \|y - x\| < r \}.$$

Definition 3 (Limit of a sqeuence)**.** Let $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that x_n converges to $\bar{x}$, and write $x_n \to \bar{x}$ if for every $\varepsilon > 0$ there exists an $N > 0$ such that for every $k > N$, $x_k \in B(\bar{x}, \varepsilon)$, i.e. for k big enough, $\|\bar{x} - x_k\| < \varepsilon$, and we say that

$$\lim_{k \to +\infty} x_k = \bar{x}.$$

Remark 4. The limit in $\mathbb{R}^n$ is the pointwise limit, so $x_k \to \bar{x}$ if and only if $(x_k)_j \to (\bar{x})_j$ for any $1 \leq j \leq n$.

Example 5. Consider the sequence $\{x_k\}_{k \in \mathbb{N}}$ defined as

$$x_k = \left(\frac{1}{k}, \frac{1}{k} \sin(k) \right) \to (0, 0)$$

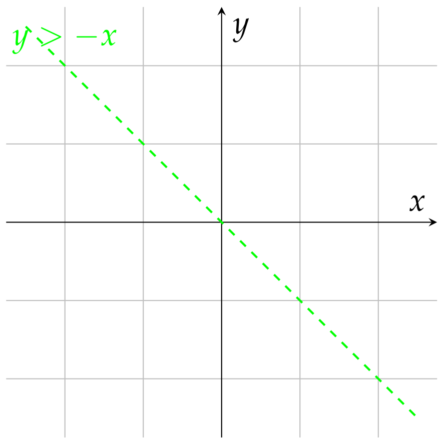

Figure 1.1: Ball of radius r centered in x on $\mathbb{R}^2$.

as $k \to +\infty$.

Definition 6 (Open/closed set)**.** Let $A \subseteq \mathbb{R}^n$. We say that A is *open* if for every $x \in A$ there exists an $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We say that A is *closed* if $\mathbb{R}^n \setminus A$ is open.

Example 7. Consider the set

$$A = \left\{(x_1, x_2) \in \mathbb{R}^2 : |x_1| < 1, |x_2| < 1\right\}.$$

We can easily verify that A is open. Observe that a set can be neither open nor closed. The sets $\mathbb{R}^n$ and $\emptyset$ are both open and closed at the same time.

Definition 8 (Neighbour)**.** We say that, for a given $x \in \mathbb{R}^n$, a subset $A \subseteq \mathbb{R}^n$ is a *neighbour* of x if A is an open set containing x.

Definition 9 (Interior, exterior, boundary, closure)**.** Let $A \subseteq \mathbb{R}^n$, $x \in \mathbb{R}^n$. We say that x is:

- an *interior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We denote the set of all interior points of A with A°;
- A *boundary point* for A if $\forall \varepsilon > 0$, $B(x, \varepsilon) \cap A \neq \emptyset$ and $B(x, \varepsilon) \cap A^C \neq \emptyset$. We denote the set of all the boundary points with ∂A;
- an *exterior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A^C$.

We denote the *closure* of A as $\bar{A} = A \cup \partial A$.

Proposition 10. *Let $\{A_k\}_{k \in \mathbb{N}}$ a sequence of open sets. Then:*

i) *the countable union of open sets is open, i.e.*

$$\bigcup_{k\in\mathbb{N}} A_k \text{ is open;}$$

ii) *the* finite intersection *of open sets is open, i.e. for* $N \in \mathbb{N}$,

$$\bigcap_{i=1}^{N} A_i \text{ is open.}$$

Proof. We prove *i*) and *ii*) separately.

i): set $E = \bigcup_{k=1}^{+\infty} A_k$. Given $x \in A$ then there exits $m > 0$ such that $x \in A_m$. Since A_m is open, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A_m \subseteq E$, so we found an open ball containing x fully contained in E. Since the choice of x is arbitrary, this must hold for every $x \in E$, hence E is open.

ii): set $E = \bigcap_{k=1}^{N} A_k$. Let $x \in E$, then $x \in A_k$ for every $1 \leq k \leq N$, hence there exists a sequence $\varepsilon_k > 0$ such that $B(x, \varepsilon_k) \subseteq A_k$ for every $k < N$. Consider $\bar{\varepsilon} = \frac{1}{2} \min_k \varepsilon_k > 0$. Then $B(x, \bar{\varepsilon}) \subseteq B(x, \bar{\varepsilon})$ for every k, hence $B(x, \bar{\varepsilon}) \subseteq E$. Again, since the choice of x is arbitrary, this must hold true for every $x \in E$, hence E is open.

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Proposition 11 (Characterization of closed sets). *Let* $C \subseteq \mathbb{R}^n$. *Then* C *is closed* $\iff$ *for every sequence* $\{x_k\}_{k\in\mathbb{N}} \subseteq C$ *such that* $x_k \to \bar{x}$, *then* $\bar{x} \in C$.

Proof. We prove $\implies$ and $\impliedby$.

" $\implies$ **":** let $C \subseteq \mathbb{R}^n$ be closed. Consider $\{x_k\}_{k\in\mathbb{N}}$ such that $x_k \to \bar{x} \in \mathbb{R}^n$, we wanto to show that $\bar{x} \in C$. Assume, by contraddiction, that $\bar{x} \notin C \implies \bar{x} \in \mathbb{R}^n \setminus C$, which is an open set, so there exists $\varepsilon > 0$ such that $B(\bar{x}, \varepsilon) \subseteq \mathbb{R}^n \setminus C \implies B(\bar{x}, \varepsilon) \cap C = \emptyset$, but $x_k \to \bar{x}$, so, for this ε, there must be an $N > 0$ such that $x_k \in B(\bar{x}, \varepsilon)$ for $k > N$, which is in contraddiction with the fact that the sequence takes values in C.

" $\Longleftarrow$ ": let C be a set. Consider $\{x_k\}_{k\in\mathbb{N}} \subseteq C$ such that $x_k \to \bar{x} \in C$. Assume, by contraddiction, C not closed. Then, for every $\varepsilon > 0$, there exists $z \in \mathbb{R}^n \setminus C$ such that $B(z,\varepsilon) \cap (\mathbb{R}^n \setminus C) = \emptyset$ and $B(z,\varepsilon) \cap C \neq \emptyset$. So we can find a sequence $\{x_k\}_{k\in\mathbb{N}}$ with values in C such that $x_k \to z$ by fixing a sequence $\{\varepsilon_k\}_{k\in\mathbb{N}}$ such that $\varepsilon_k \searrow 0$, so $x_k \in B(z,\varepsilon_k) \cap C$, $x_k \to z$, so, by assumption $z \in C$, which is in contraddiction with $z \in \mathbb{R}^n \setminus C$.

■

Definition 12 (Bounded/finite set)**.** A set $A \subseteq \mathbb{R}^n$ is *bounded* if there exists $R > 0$ such that $A \subseteq B(0,R)$.

The set A is *finite* if $A = \{x_1, \dots, x_m\} \subseteq \mathbb{R}^n$.

Definition 13 (Continuity)**.** Let $f : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $\bar{x} \in E$. We say that f is *continous* in $\bar{x}$ if for every sequence $\{x_k\}_{k\in\mathbb{N}}$ such that $x_k \to \bar{x}$ in $\mathbb{R}^n$, then $f(x_k) \to f(\bar{x})$ in $\mathbb{R}^m$.

Definition 14 (Cluster/isolated point)**.** Let $E \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that $\bar{x}$ is a *cluser point* for E if there exists a sequence $\{x_k\}_{k\in\mathbb{N}}$ such that, for every $\bar{x} \neq x_k \to \bar{x}$. We denote tha set of all cluster points with $\mathrm{Acc}(E)$.

A point $\bar{x}$ is *isolated* in E if $x \in E \setminus \mathrm{Acc}(E)$.

Definition 15 (Limit)**.** Let $f : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $\bar{x} \in \mathrm{Acc}(E)$. Fixed $\ell \in \mathbb{R}^m$, we say that

$$\lim_{x\to\bar{x}} f(x) = \ell$$

if for every sequence $\{x_k\}_{k\in\mathbb{N}}$, $x_k \to \bar{x}$, $x_k \neq \bar{x}$ for every k, we have $f(x_k) \to \ell$. Notice that we allow $\bar{x}$ to not necessarely be in E.

Example 16. Consider the function $f : \{0\} \cup [1,2] \subset \mathbb{R} \to \mathbb{R}$, $x \mapsto x$. Then it has no meaning to speak about $\lim_{x\to 0} f(x)$, as $0 \notin \mathrm{Acc}(\{0\} \cup [1,2])$, while f is continous at 0, since the only sequence converging to 0 in $\{0\} \cup [1,2]$ is the sequence which is identically 0.

Remark 17. If $f : \mathbb{R}^n \to \mathbb{R}$ is continous, then:

- $\{x : f(x) < c\}$ is *open* $\forall c \in \mathbb{R}$;
- $\{x : f(x) \leq c\}$ is *closed* $\forall c \in \mathbb{R}$;
- $\{x : f(x) = c\}$ is *closed* $\forall c \in \mathbb{R}$.

Chapter 2

Lecture 2 of 04/03/2026

Our main focus from now on is to solve the problem

$$\inf_{x \in A} f(x)$$

for some $A \subseteq \mathbb{R}^n$ and some $f : \mathbb{R}^n \to \mathbb{R}$. We will restrict ourselfs only to certain functions and sets.

Definition 1 (Infimum/supremum)**.** Let $E \subseteq \mathbb{R}$. We denote with $\inf E =$ largest $z \in \mathbb{R}$ such that $z \leq x$ for every $x \in E$. By convention we set $\inf \emptyset = +\infty$.

Similarly, we denote with $\sup E$ the smallest $z \in \mathbb{R}$ such that $z \geq x$ for every $x \in E$, with the convention $\sup \emptyset = -\infty$. More formally,

$$z = \inf E \quad \text{if } z \leq x \, \forall x \in E \text{ and } \forall \varepsilon > 0 \, \exists y \in E \text{ s.t. } y \leq z + \varepsilon$$
$$z = \sup E \quad \text{if } z \geq x \, \forall x \in E \text{ and } \forall \varepsilon > 0 \, \exists y \in E \text{ s.t. } y \geq z - \varepsilon$$

Remark 2. The inf and sup aren't necessarely inside the set. If they are, we speak of min and max.

Axiom 1 — *Completeness of* $\mathbb{R}$

Every bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$ admits a supremum in $\mathbb{R}$.

Theorem 3 (Bolzano-Weierstrass)**.** *Any bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$ admits a cluster point, i.e. there exists $\bar{x} \in \mathbb{R}$ and a subsequence $\left\{x_{n_j}\right\}_{j \in \mathbb{N}}$ such that $x_{n_j} \to \bar{x}$ as $j \to +\infty$.*

Proof. Since $\{x_n\}_n$ is bounded, there exists $R > 0$ such that $x_n \in [-R, R]$ for every $n \in \mathbb{N}$. Set $I_1 = [-R, R]$, select x_{n_1} a value in I_1. Divide I_1 in half and select the one subinterval containing inifitely many x_ns, call it $I_2 = [a_2, b_2]$ (here $a_2 = -R, b_2 = 0$ or $a_2 = 0, b_2 = R$). We have $I_2 \subset I_1$, $|I_2| = |I_1|/2$, and choose x_{n_2} a value in I_2. Repeat this process to cut and select. We end up, at the k-th step, with a sequence of subintervals $I_1, I_2, \ldots, I_k$ and 3 subsequences: $x_{n_1}, x_{n_2}, \ldots, x_{n_k}$

the sequence of selected values, $a_1, a_2, \dots, a_k$ the sequence of left boundaries and $b_1, b_2, \dots, b_k$ the sequence of right boundaries.

We observe that $a_{j+1} \geq a_j$ and $b_{j+1} \leq b_j$ for all $1 \leq j \leq k$. Moreover, $a_j \leq R$ and $b_j \geq -R$ for all j, so, by the axiom of completeness of $\mathbb{R}$,

- there exists $\bar{a} \in \mathbb{R}$ such that $a_k \to \bar{a} = \sup_k a_k$
- there exists $\bar{b} \in \mathbb{R}$ such that $b_k \to \bar{b} = \inf_k b_k$

We observe that $\bar{a} = \bar{b}$. Indeed, $\left|\bar{b} - \bar{a}\right| = \left|\bar{b} - b_k + b_k - a_k + a_k - \bar{a}\right| \leq \left|\bar{b} - b_k\right| + |\bar{a} - a_k| + |b_k - a_k|$. The terms $\left|\bar{b} - b_k\right|$ and $|\bar{a} - a_k|$ goes to 0 as $k \to +\infty$ for what we observed before. Moreover, $|b_k - a_k| = 2R/2^{k-1} \to 0$ as $k \to +\infty$.

Set $\bar{x} = \bar{a} (= \bar{b})$. Then the subsequence $\left\{x_{n_j}\right\}_j$ is converging to $\bar{x}$: indeed $\left|x_{n_j} - \bar{x}\right| = \left|x_{n_j} - a_j + a_j - \bar{x}\right| \leq \left|a_{n_j} - a_j\right| + \left|a_j - \bar{x}\right|$.

Since $a_j \to \bar{x} = \bar{a}$, the term $\left|a_j - \bar{x}\right| \to 0$. Notice that, by construction, $x_{n_j} \in I_j$, so $\left|x_{n_j} - a_j\right| \leq \left|I_j\right| \to 0$.

■

Definition 4. Let $f : \mathbb{R}^n \to \mathbb{R}$. We say that $\bar{x} \in \mathbb{R}^n$ is a *local maximum* for f if there exists a neighbourhood U of $\bar{x}$ such that $f(x) \leq f(\bar{x})$ for every $x \in U$. A *local strict maximum* has the strict inequality.

Similarly, $\bar{x}$ is *local minumum* for f if $\exists U$ neighbourhood of $\bar{x}$ such that $f(x) \geq f(\bar{x})$ for all $x \in U$.

If there exists $\bar{x}$ such that $f(x) = \inf_{x \in A} f(x)$, the $\bar{x}$ is a *global minimum* over A.

Definition 5. We denote with $\arg_{x \in A} \min f(x)$ the set of minumum points for f over A, i.e

$$\arg_{x \in A} \min f(x) = \left\{\bar{x} \in \mathbb{R}^n : f(\bar{x}) = \inf_{x \in A} f(x)\right\}.$$

Theorem 6 (Weierstrass). *Let $A \subseteq \mathbb{R}^n$ be a compact set, $f : A \to \mathbb{R}$ a continous function. The there exists x_m, $x_M \in A$ such that*

$$f(x_m) = \min_{x \in A} f(x)$$
$$f(x_M) = \max_{x \in A} f(x).$$

Proof. We prove the statement for the minimum point.

Let $\ell = \inf_{x \in A} f(x)$. We can always find a sequence $\{x_n\}_n \subseteq A$ such that $f(x_n) \to \ell$ by the continuity of f. Since A is bounded, $\{x_n\}_n$ is bounded as well. By Bolzano-Weierstrass, there exists a subsequence $\{x_{n_k}\}_k$ converging to $\bar{x} \in A$ by closerness of A. By continuity, in particular f is continous at $\bar{x}$, ■

Remark 7. For $f : \mathbb{R}^n \to \mathbb{R}$, we define the *gradient* of f as

$$\nabla f(c) = \left(\frac{\partial f}{\partial x_1}(c), \dots, \frac{\partial f}{\partial x_n}(c) \right)$$

the vector of partial derivatives, defined as

$$\frac{\partial f}{\partial x_i}(c) = \lim_{h \to 0} \frac{f(c + he_i) - f(c)}{h}$$

whenever this limit exists finite, and where e_i is the i-th vector of the canonical base of $\mathbb{R}^n$. Furthermore, we say that f is *differentiable* at c if

$$f(x) = f(c) + \langle \nabla f(c), x - c \rangle + \mathcal{O}(\|x - c\|^2)$$

for $x \to c$.

We now set ouf focus on convex analysis.

Definition 8 (Convex set)**.** Let $C \subseteq \mathbb{R}^n$. We say that C is *convex* if, for any $x, y \in C$, and for any $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \in C$.

Proposition 9. *a)* $\bigcap_{i \in I} C_i$ *is convex if* C_i *is convex for any* $i \in I$ *finite or countable;*

b) the algebraic sum $C_1 + C_2 = \{x + y : x \in C_1, y \in C_2\}$ *is convex if* C_1 *and* C_2 *are convex;*

c) the set $\lambda C = \{\lambda x : x \in C\}$ *is convex if* C *is convex for any* $\lambda \in \mathbb{R}$*;*

d) $\overline{C}, C^\circ$ *are convex if* C *is convex;*

e) if a function $f : C \to \mathbb{R}$ *is affine, then the set* $f(C) = \{y : y = f(x), x \in C\}$ *is convex if* C *is convex.*

Proof. We prove all the propositions separately.

a): suppose C_i convex and let $C = \bigcap_{i \in I} C_i$. Let $x, y \in C$, $\alpha \in [0,1]$. Since $x, y \in C$, in particular $x, y \in C_i$ for all $i \in I$. But then also $\alpha x + (1-\alpha)y \in C_i$ for all $i \in I$, hence C is convex.

b): let $x = x_1 + x_2 \in C_1 + C_2$, $x_1 \in C_1$ and $x_2 \in C_2$, take $y \in C_1 + C_2$ of the same form.

$$\alpha x + (1 - \alpha y) = \underbrace{\alpha x_1 + (1-\alpha)y_1}_{\in C_1} + \underbrace{\alpha x_2 + (1-\alpha)y_2}_{\in C_2} \in C.$$

c): trivial.

d): consider $\{x_k\}_k \subseteq C$ and $\{y_k\}_k \subseteq C$ such that $x_k \to x$ and $y_k \to y$. Then the sequence $\{\alpha x_k + (1-\alpha)y_k\}_k \subseteq C$. But then also $\alpha x + (1-\alpha)y \in C$ as limit of sequences valued in C.

Let now $x, y \in C^\circ$. Then there exists $r > 0$ such that $B(x,r) \subseteq C$ and $B(y,r) \subseteq C$. Consider $\alpha x + (1-\alpha)y$. Our claim is: $B(\alpha x + (1-\alpha)y, r) \subseteq C$. Indeed, any $z \in B(\alpha x + (1-\alpha)y, r)$ is of the form $z = \alpha x + (1-\alpha)y + w$ with $\|w\| < r$, so $\alpha(x + w) + (1-\alpha)(y + w) \in C$ by convexity of C, hence the claim.

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Chapter 3

Lecture 3 of 10/03/2026

Example 1 (Convex sets)**.** The following sets are convex in $\mathbb{R}^n$:

- **hyperplanes**: $\left\{x \in \mathbb{R}^n : a^\top x = b\right\}$ for $a \in \mathbb{R}^n$, $b \in \mathbb{R}$;
- **halfspaces**: $\left\{x \in \mathbb{R}^n : a^\top x \leq b\right\}$ for $a \in \mathbb{R}^n$, $b \in \mathbb{R}$;
- **polyhedra**: $\left\{x \in \mathbb{R}^n : a_i^\top x \leq b_i\right\}$ for $a_i \in \mathbb{R}^n$, $b_i \in \mathbb{R}$ for $i = 1, \ldots, m$ (they are intersecttion of half spaces).

Half cones are convex as well.

Definition 2 (Convex/concave function)**.** Let $C \subseteq \mathbb{R}^n$ be convex. Consider $f : C \to \mathbb{R}$. We say that f is *convex* if the following holds:

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y) \quad \forall x, y \in C,\ \forall \alpha \in [0,1].$$

We say that f is *concave* if $-f$ is convex.

Example 3. Affine functions are convex. Indeed, let $f(x) = a^\top x + b$, $a \in \mathbb{R}^n$. Indeed,

$$\begin{aligned} f(\alpha x + (1-\alpha)y) &= \langle a, \alpha x + (1-\alpha)y\rangle + b(1-\alpha+\alpha) \\ &= \alpha\langle a, x\rangle + \alpha b + (1-\alpha)\langle a, y\rangle + (1-\alpha)b \\ &= \alpha f(x) + (1-\alpha) f(y). \end{aligned}$$

Example 4. The norm in $\mathbb{R}^n$ is convex, as an easy consequence of triangular inequality.

We can observe that f convex implies that the set $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ is convex for all $\gamma \in \mathbb{R}$. Indeed, for every $x, y \in V_\gamma$ we have $f(x) \leq \gamma$ and $f(y) \leq \gamma$. We see that $f(\alpha x + (1-\alpha)y) \in V - \gamma$. Indeed, by convexity of f we have $f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y) = \alpha\gamma + (1-\alpha)\gamma = \gamma$.

In general, the converse is not true, i.e. convex level sets *does not imply* f convex.

From now on, $f : \mathbb{R}^n \to \overline{\mathbb{R}}$, where $\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$.

Definition 5 (Epigraph)**.** Let $f : \mathbb{R}^n \to \overline{\mathbb{R}}$. We define the set

$$\operatorname{epi}(f) = \left\{(x, w) \in \mathbb{R}^{n+1} : x \in X,\, w \in \mathbb{R},\, f(x) \leq w\right\}.$$

An intuitive definition could be "*the set of points above the graph of f*".

Definition 6 (Effective domain)**.** Given $f : X \subseteq \mathbb{R}^n \to \overline{\mathbb{R}}$ we define the *effective domain* of f as the set

$$\operatorname{dom}(f) = \{x \in X : f(x) < +\infty\} = \{x \in X : \exists\, w \in \mathbb{R} \text{ s.t. } (x, w) \in \operatorname{epi}(f)\}.$$

Definition 7 (Proper function)**.** A function $f : X \subseteq \mathbb{R}^n \to \overline{\mathbb{R}}$ is said *proper* if $f(x) < +\infty$ and $f(x) > -\infty$ for all $x \in X$. In other words, $\operatorname{epi}(f)$ does not contain any vertical line.

Definition 8 (Extension of convex functions)**.** Let $C \subseteq \mathbb{R}^n$ be a convex set. A funciton $f : C \to \overline{\mathbb{R}}$ is *convex* if $\operatorname{epi}(f)$ is convex in $\mathbb{R}^{n+1}$.

Remark 9. If $f : C \to \mathbb{R}$ the definitions coincides.

Proof. Assume $\operatorname{epi}(f)$ convex in $\mathbb{R}^{n+1}$, then the points $(x, f(x)), (y, f(y)) \in \operatorname{epi}(f)$, but also $\alpha(x, f(x)) + (1-\alpha)(y, f(y)) \in \operatorname{epi}(f)$. This implies $(\alpha x + (1-\alpha)y, \alpha f(x) + (1-\alpha)f(y)) \in \operatorname{epi}(f) \implies f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$.

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Definition 10 (Convex indicator function)**.** Given $A \subseteq \mathbb{R}^n$, we define the *convex indicator function* of the set A as

$$\delta_A(x) = \begin{cases} 0 & \text{if } x \in A \\ +\infty & \text{otherwise} \end{cases}$$

This is coherent with the infimum operation, as

$$\inf_{x \in A} f(x) = \inf_{x \in \mathbb{R}^n} f(x) + \delta_A(x).$$

Definition 11 (Liminf)**.** Let $f : x \subseteq \mathbb{R}^n \to \mathbb{R}$, $c \in \operatorname{Acc}(X)$. We say that

$$\ell = \liminf_{x \to c} f(x)$$

if $\ell = \inf_{x_k} \lim_{k \to +\infty} f(x_k)$ where $\{x_k\}_k$ is a sequence in X, $\{x_k\}_k \to c$, $x_k \neq c$ for all k and $\lim_{k \to \infty} f(x_k)$ exists finite.

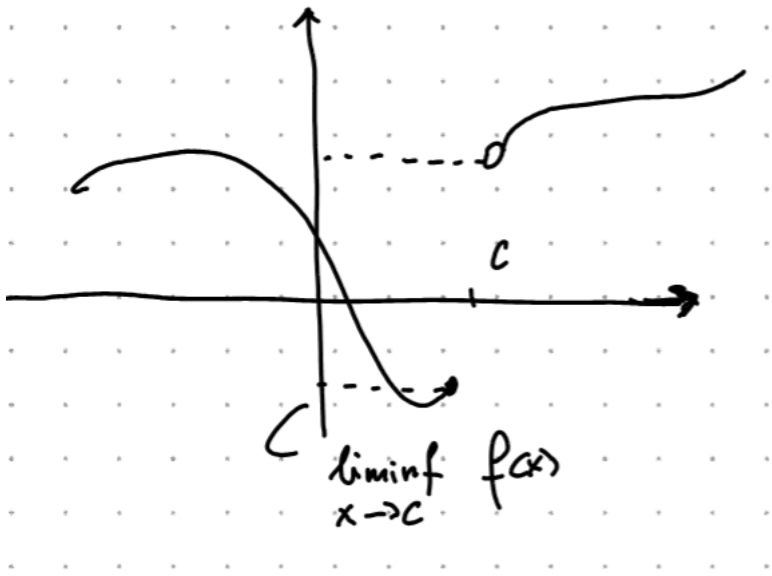


Figure 3.1: The liminf of f at $x = 1$ is 1, even if $f(1) = 0.5$.

Definition 12 (Lower semicontinuity). Let $f : X \subseteq \mathbb{R}^n \to \mathbb{R}$. We say that f is *lower semicontinuous* (lsc for short) at $c \in X$ if

$$f(c) \leq \liminf_{x \to c} f(x).$$

Theorem 13 (Weierstrass). *The Weierstrass theorem still allows a minimum point for a lsc funciton.*

Proposition 14. *Let $f : \mathbb{R}^n \to \overline{\mathbb{R}}$. The following are equivalent:*

i) The level sets $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ are closed for all $\gamma \in \mathbb{R}$;

ii) f is lsc;

iii) epi (f) *is closed in $\mathbb{R}^{n+1}$.*

Proof. We prove $i) \implies ii) \implies iii) \implies i)$.

$i) \implies ii)$**:** suppose V_γ closed for all γ. Assume by contraddiction f not lsc, so there exists $c \in \mathbb{R}^n$ s.t. $f(c) > \liminf_{x \to c} f(x)$. This implies that there exists a sequence $\{x_k\}_k$, $x_k \to c$ and a $\gamma \in \mathbb{R}$ s.t. $f(c) > \gamma \geq \lim_{k \to +\infty} \inf f(x_k)$. For k big enough , we have $f(x_k) \leq \gamma$, so $x_k \in V_\gamma$. By closerness of V_γ, it must be $c \in V_\gamma \implies f(c) \leq \gamma$, which is in contraddiction with $f(c) > \gamma$.

ii) $\implies$ *iii)*: assume f lsc, we want to show that any sequence $\{(x_k, w_k)\}_k \subseteq \mathrm{epi}\,(f)$ converging to $(\bar{x}, \bar{w})$ must satisfy $(x, w) \in \mathrm{epi}\,(f)$. Assime $x_k \to \bar{x}$, $w_k \to \bar{w}$. Then $f(x_k) \leq w_k$ for all k. Since $x_k \to \bar{x}$ and f is lsc, it must be that $f(\bar{x}) \leq \liminf_{k\to+\infty} f(x_k) \leq \liminf_{k\to+\infty} w_k = \bar{w}$, hence $(\bar{x}, \bar{w}) \in \mathrm{epi}\,(f)$.

iii $\implies$ *i)*: assume $\mathrm{epi}\,(f)$ closed. Let $\gamma \in \mathbb{R}$, $V_\gamma = \{x : f(x) \leq \gamma\}$. Consider the sequence $\{x_k\}_k \subseteq V_\gamma$ s.t $x_k \to \bar{x}$. We want to show $\bar{x} \in V_\gamma$. The points $(x_k, \gamma) \in \mathrm{epi}\,(f)$ for all k, so, by closerness of $\mathrm{epi}\,(f)$, $(\bar{x}, \gamma) \in \mathrm{epi}\,(f)$. This impies $f(\bar{x}) \leq \gamma$, hence $\bar{x} \in V_\gamma$.

■

Proposition 15. *Let $f : \mathbb{R}^m \to (-\infty, +infty]$, $A \in \mathbb{R}^{m\times n}$, $F : \mathbb{R}^n \to (-\infty, +\infty]$, $x \mapsto f(Ax)$. If f is convex, then F is convex. If f is closed, then F is closed.*

Proof. Assume f convex. Then

$$F(\alpha x + (1-\alpha)y) = f(\alpha Ax + (1-\alpha)Ay) \leq \alpha f(Ax) + (1-\alpha)f(Ay).$$

Assume f closed, so $\mathrm{epi}\,(f)$ is closed and f is lsc. Take a sequence $x_k \to \bar{x}$. Then $\liminf_k F(x_k) = \liminf_k f(Ax_k)$ Let $y_k = Ax_k$, so $y_k \to A\bar{x}$ by continuity of the linear map Ax. This implies $\liminf_k f(Ax_k) \geq f(A\bar{x}) = F(\bar{x})$.

■

Proposition 16. *Let $\{f_i\}_{i=1}^m$ be a family of convex and closed function s.t. $f_i : \mathbb{R}^n \to (-\infty, +\infty]$, and $\gamma_1, \ldots, \gamma_m > 0$. Then the function defined by $F(x) = \sum_{i=1}^m \gamma_i f_i(x)$ is closed and convex.*

Proposition 17. *Let $f_i : \mathbb{R}^n \to (-\infty, +\infty]$ for $i \in I$ any countable index set. Let $f(x) = \sup_i f_i(x)$. If f_i is closed convex for all $i \in I$ then f is closed convex.*

Proof. Let $(x, w) \in \mathrm{epi}\,(f)$, then $f(x) \leq w$, so $f_i(x) \leq w$ for all i. This implies $(x, w) \in \mathrm{epi}\,(f_i)$ for all i, hence $\mathrm{epi}\,(f) = \bigcap_{i\in I} \mathrm{epi}\,(f_i)$, which is convex as countable intersection of convex sets. The same argument holds for closerness.

■

Chapter 4

Lecture 4 of 11/03/2026

We ask ourselfs what happens if f is convex and we add some regularity hypotesis. We first remark that if f is differentiable at c, near this point, the function looks like an hyperplane. The intuition is: function must be always above its tangent hyperplane.

Proposition 1. *Let $C \subseteq \mathbb{R}^n$ be convex, $A \subseteq \mathbb{R}^n$ open s.t. $C \subseteq A$. Let $f : A \to \mathbb{R}$ convex and differentiable. Then*

i) f is convex on $C \iff f(z) \geq f(x) + \langle \nabla f(x), z - x \rangle$ for all $x, z \in C$

ii) f is strictly convex over $C \iff f(z) > f(x) + \langle \nabla f(x), z - x \rangle$ for all $x, z \in C$ s.t. $x \neq z$

Proof. Let f be convex on C. Let $x, z \in C$. Observe that

$$\lim_{h \to 0^+} \frac{f(x + h(z - x)) - f(x)}{h} = \frac{\partial}{\partial h} f(x + h(z - x))|_{h=0} = \nabla f(x) \cdot (z - x).$$

Then

$$\begin{aligned} f(x) + \langle \nabla f(x), z - x \rangle &= f(x) + \lim_{h \to 0^+} \frac{f(x + h(z - x)) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{hf(x) - f(x) + f(hz + (1 - h)x)}{h} \\ &\leq \lim_{h \to 0} \frac{-(1 - h)f(x) + hf(z) - (1 - h)f(x)}{x} \\ &= \lim_{h \to 0} f(z) = f(z). \end{aligned}$$

Let now $\alpha \in [0, 1]$, $x, y \in C$, $z = \alpha x + (1 - \alpha)y$.

$$\begin{aligned} f(x) &\geq f(z) + \langle \nabla f(z), x - z \rangle \cdot \alpha \\ f(y) &\geq f(z) + \langle \nabla f(z), y - z \rangle \cdot (1 - \alpha) \end{aligned}$$

Take the sum of the above quantities, then

$$\alpha f(x) + (1-\alpha) f(y) \geq \alpha f(z) + (1-\alpha) f(z) + \alpha \langle \nabla f(z), x - z \rangle + (1-\alpha) \langle \nabla f(z), y - z \rangle$$

hence, by linearity of the scalar product,

$$\alpha f(x) + (1-\alpha) f(y) \geq f(z) + \langle \nabla f(z), \alpha x + (1-\alpha) y - z \rangle.$$

Hence, f is convex.

For the strict convexity, we repeat the same process with the strict inequality.

■

If at some point x^* we have $\langle \nabla f(x^*), z - x^* \rangle \geq 0$ for all z, then x^* is a minimum point. This is also a necesary condition: assume x^* minumum point for f over C and, by contraddiction, thath there exists $z \in C$ s.t. $\langle \nabla f(x^*), z - x^* \rangle < 0$, but then

$$\frac{\partial}{\partial h} f(x^* + h(z - x^*))|_{h=0} = \langle \nabla f(x^*), z - x^* \rangle < 0$$

so f decreases in the direction $z - x^*$, which contrdaddicts the fact that x^* is a minimum.

We summarize this fact in the following proposition, which proof what we just did.

Proposition 2. *Let $C \subseteq \mathbb{R}^n$ be convex, $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable on a convex open set $A \subseteq C$. Then a point $x^* \in C$ is a minimum for f if and only if*

$$\langle \nabla f(x^*), z - x^* \rangle \geq 0 \quad \forall z \in C.$$

Theorem 3 (Projection theorem)**.** *Let C be a nonempty convex closed sunset of $\mathbb{R}^n$, $z \in \mathbb{R}^n$. Then there exists a unique vector $x^* \in C$ (the projection of z on C) that minimizes the function $f(x) = \|x - z\|$ over C.*

Furthermore, the point x^* is the projection of z onto C if and only if

$$\langle z - x^*, x - x^* \rangle \leq 0 \quad \forall x \in C.$$

Proof. Optimizing the function $f(x) = \|z - x\|$ is the same as optimizing the funciton $f(x) = \frac{1}{2}\|z - x\|^2$. Then f is now differentiable, and $\nabla f(x) = z - x$.

$$\begin{aligned} x^* \text{ is minimum} &\iff \langle \nabla f(x^*), x - x^* \rangle \geq 0 \quad \forall x \in C \\ &\iff \langle -x^* + z, x - x^* \rangle \leq 0. \end{aligned}$$

Does such an x^* exists? We observe that

$$\min_{x \in C} f(x) = \min_{x \in C \cap \{y : \|y - z\| \leq \|z - w\|\}, w \in C} f(x).$$

We notice that we are optimizing a continous function on a closed set, as we are optimizing on $C \cap \overline{B(z, \|z - w\|)}$, so, by Weierstrass there exists a minimum. This is also unique, indeed, let x_1^*, x_2^* be two minima, then

$$\langle z - x_1^*, x_2^* - x^*1 \rangle \leq 0 \quad \forall z \in C$$
$$\langle z - x_2^*, x_2^* - x^*1 \rangle \leq 0 \quad \forall z \in C$$

Taking the sum ad using the linearity of the scalar product, we obtain

$$\langle x_1^*, x_2^*, z - x_2^* - z + x_1^* \rangle \leq 0 \implies \|x_1^* - x_2^*\| \leq 0 \implies x_1^* = x_2^*.$$

■

Convex hulls

The idea is ton"convexify" sets that are not convex. Let X be a set. The *convex hull* of X conv $(()\, X)$ = intersection of all convex sets containing X. A convex combination of elements of X is a vector in the form $\sum_{i=1}^m \alpha_i x_i$, where x_i are points in X, $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$.

Remark 4. If X is convex, any convex combination of elements of X belongs to X.

Remark 5.

$$\operatorname{conv}(X) = \left\{ z = \sum_{i=1}^m \alpha_i x_i : \alpha_i > 0,\ x_i \in X,\ \sum_i \alpha_i = 1,\ \forall m > 0 \right\}.$$

Remark 6. If $A : \mathbb{R}^n \to \mathbb{R}^n$ is a linear tranformation, then $\operatorname{conv}(Ax) = A \operatorname{conv}(X)$.

Remark 7. $\operatorname{conv}\left(\sum_i X_i \right) = \sum_i \operatorname{conv}(X_i)$.

Definition 8 (Affine hull)**.** Let $X \subseteq \mathbb{R}^n$. The *affine hull* of X, denoted with $\operatorname{aff}(X)$ is the intersection of all affine sets containing X.

Recall that M is an affine set if M is in the form $M = \bar{x} + S$, where S is a linear subspace of $\mathbb{R}^n$.

Notice that $\operatorname{conv}(X) \subseteq \operatorname{aff}(X)$, and intersection of affine spaces is still an affine space, so $\operatorname{aff}(X)$ is an affine space, so $affX = \bar{x} + S$ for sone $\bar{x} \in \mathbb{R}^n$ and S subspace. We define the dimension of $\operatorname{aff}(X)$ as the dimension of S. S is also called the parallel subspace of the affine space.

Remark 9. $\operatorname{aff}(X) = \operatorname{aff}(\operatorname{conv}(X)) = \operatorname{aff}(\operatorname{cl}(X))$ where $\operatorname{cl}(X)$ denotes the closure of X.

Definition 10 (Cone generated by X)**.** Let $X \subseteq \mathbb{R}^n$. We define the *cone generated by* X the set

$$\operatorname{cone}(X) = \left\{ \sum_{i=1}^{m} \alpha_i x_i : \alpha_i \geq 0,\, x_i \in X,\, m = 0 \right\}.$$

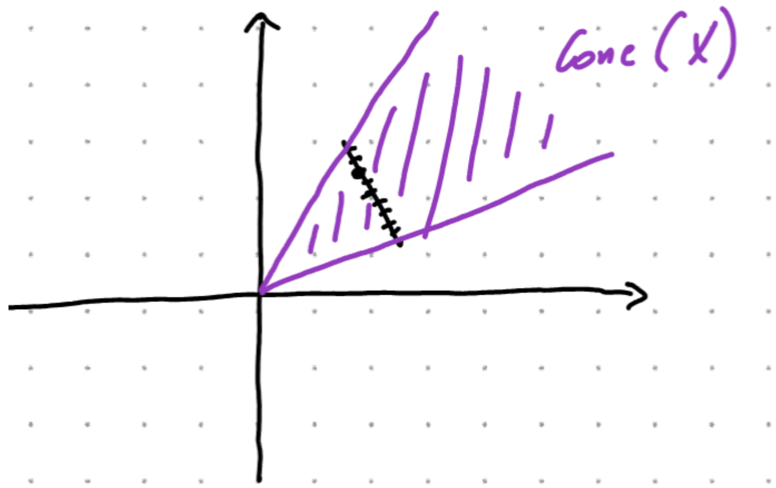


Figure 4.1: X (black) and cone generated by X (purple)

Chapter 5

Lecture 5 of 17/03/2026

To be done.

CHAPTER 6

LECTURE 6 OF 18/03/2026

THEOREM 1 (Charateodory). *Let $X \subseteq \mathbb{R}^n$ be nonempty. Then*

a) every nonzero vector from $\operatorname{cone}(X)$ *can be represented as a positive combination of linearly independent vectors in X;*

b) every vector fron $\operatorname{conv}(X)$ *can be represented as a convex combination of no more then $n+1$ vectors in X.*

Proof. We prove a) and b) separately.

a): Let $y \in \operatorname{cone}(X) \implies y = \sum i = 1^m \alpha_i x_i$, where $\alpha_i > 0$, $x_i \in X$. We can assume m to be the smallest index such that this representation holds true. We have to prove that $x_1, \ldots, x_m$ to be linearly independent. By contraddiction, assume them dependent, so there exists $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$ such that $\sum_{i=1}^{m} \lambda_i x_i = 0$. Without loss of generality, assume at least one $\lambda_i > 0$. Define $\bar{\gamma} = \max \{\gamma : \alpha_i - \gamma\lambda_i \geq 0\} < +\infty$. Observe that we can write $y = \sum_i \alpha_i x_i + \bar{\gamma} \underbrace{\sum_i \lambda_i x_i}_{=0} = \sum_{i=1}^{m} (\alpha_i - \bar{\gamma}\lambda_i) x_i$. Then at leas one of the last coefficents is zero, which is in contraddiction with the minimality of m.

b): Consider the set $Y = \{(y, 1) : y \in X\} \subseteq \mathbb{R}^{n+1}$. Let $x \in \operatorname{conv}(X) \implies x = \sum_{i=1}^{m} \gamma_i x_i, \sum_i \gamma_i = 1$. The point $(x, 1) \in \operatorname{cone}(Y) \implies \sum_i \gamma_i (x_i, 1) = (x, 1)$. By point a), $(x, 1) = \sum_{i=1}^{m} \alpha_i (\tilde{x}_i, 1)$, with $(\tilde{x}_i, 1)$ linearly independent in Y. Hence $x = \sum_{i=1}^{m} \alpha_i \tilde{x}_i$ with $\sum_i \alpha_i = 1$. From the fact that $(\tilde{x}_i, 1)$ are independent in Y, we conclude that $m \leq n + 1$.

■

Proposition 2. *Let* $X \subseteq \mathbb{R}^n$ *be compact (closed and bounded). Then* $\operatorname{conv}(X)$ *is compact.*

Proof. It is easy to show that $\operatorname{conv}(X)$ is bounded. Since X is bounded, $\exists R > 0$ such that $\|x\| \leq R$ for any $x \in X$. Then $x \in \operatorname{conv}(X) \iff x = \sum_i \gamma_i x_i \implies \|x\| \leq \sum_i \gamma_i \|x_i\| \leq \sum_i \gamma_i R = R$.

Closerness: Take a sequence $\{x_k^k\}_k \subseteq \operatorname{conv}(X)$ and assume $x_k \to \bar{x}$ as $k \to +\infty$. We want to show that $\bar{x} \in \operatorname{conv}(X)$. Each x^k can be written as

$$x_k = \sum_{i=1}^{m} \alpha_i^k x_i^k$$

by the Charateodory's theorem, with $\sum_i \alpha_i = 1$, $\alpha_i \geq 0$, $x_i \in X$. Consider the sequence $\left\{(\alpha_1^k, \ldots, \alpha_{n+1}^n, x_1^k \ldots, x_{n+1}^k)\right\} \subseteq \mathbb{R}^{n+1+n(n+1)} = \mathbb{R}^{(n+1)^2}$. This is a bounded sequence and converges (up to subsequences by BW) to $(\alpha_1, \ldots, \alpha_{n+1}, x_1, \ldots, x_{n+1})$. Observe that, since $\sum_i \alpha_i^k = 1 \implies \sum_i \alpha_i = 1$, $\alpha_i \geq 0$. Observe furthermore that all $x_i \in X$ by closerness of X, so $\sum_i \alpha_i^k x_i^k \to \sum_i \alpha_i x_i$, hence $\bar{x} = \sum_{i=1}^{n+1} \alpha_i x_i$, hence $\bar{x} \in \operatorname{conv}(X)$.

∎

Relative interiors

If we consider a convex set in $\mathbb{R}^n$ which lives on an m-dimensional variety, with $m < n$ (as in 6.1), we get that the interior points of that convex sets forms the emptyset.

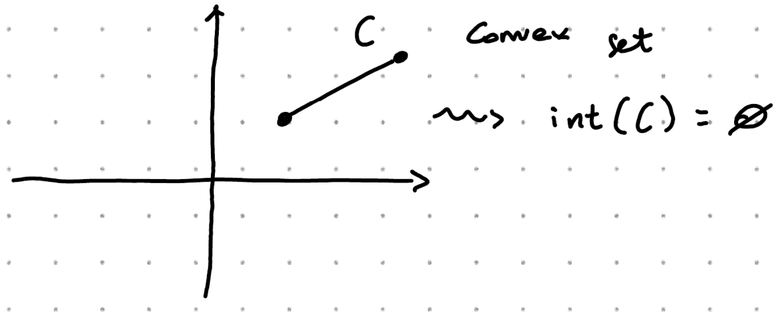


Figure 6.1: Example of emptyness of interior

Definition 3 (Relative interior point). Let $C \subseteq \mathbb{R}^n$ be a convex set, $c \in C$. We say that x is a *relative interior point* if there exists an $r > 0$ such that $B(x, r) \cap \operatorname{aff}(C) \subseteq C$.

The *relative interior* is the set of all the relative interior points, and we denote it with $\operatorname{ri}(C)$.

Furthermore we say that C is *relatively open* if $C = \operatorname{ri}(C)$.

Proposition 4 (Line-segment principle (LSP)). *Let C be a nonempty convex set. If $x \in \operatorname{ri}(C)$ and $\bar{x} \in \operatorname{cl}(C)$, then all points on the line segment connecting x and $\bar{x}$, except possibly $\bar{x}$ belongs to $\operatorname{ri}(C)$.*

Proof. Assume $\bar{x} \in C \implies x \in \operatorname{ri}(C)$, so there exists $r > 0$ such that $B(x, r) \cap \operatorname{aff}(C) \subseteq C$. Let $\alpha \in (0, 1)$ and set $x_\alpha = \alpha x + (1 - \alpha)\bar{x}$. We have to prove that $x_\alpha \in \operatorname{ri}(C)$ for all $\alpha \in (0, 1)$.

Consider $B(x_\alpha, \alpha r)$, we have to show that $B(x_\alpha, \alpha r) \cap \operatorname{aff}(C) \subseteq C$. Pick $y_\alpha \in B(x_\alpha, \alpha r) \cap \operatorname{aff}(C)$. Observing that y_α belongs both to $B(x_\alpha, \alpha r)$ and $\operatorname{aff}(C)$, in particular there exists $z_\alpha \in \mathbb{R}^n$, $\|z_\alpha\| \leq \alpha r$ such that $y_\alpha = x_\alpha + z_\alpha$, hence $y_\alpha = \alpha x + (1 - \alpha)\bar{x} + z_\alpha = \alpha \underbrace{\left(x + \frac{z_\alpha}{\alpha}\right)}_{\in B(x,r)} + \underbrace{(1 - \alpha)\bar{x}}_{\in C}$.

Observe that $x + \frac{z_\alpha}{\alpha} \in \operatorname{aff}(C)$. Indeed $x \in \operatorname{aff}(C)$. Consider $z_\alpha = y_\alpha - x_\alpha$, then $x_\alpha \in C \implies x_\alpha \in \operatorname{aff}(C)$, while $y_\alpha \in \operatorname{aff}(C)$. Putting all toghether, $z_\alpha \in$ parallel space to $\operatorname{aff}(C)$, so we remain in the affine space, as we are summing a point in the affine space and a point in the parallel subspace. Moreover $x + \frac{z_\alpha}{\alpha} \in B(x, r)$ as $\|\frac{z_\alpha}{\alpha}\| \leq \frac{\alpha r}{\alpha} = r$, hence $x + \frac{z_\alpha}{\alpha} \in C$.

Consider now the case $\bar{x} \in \mathrm{cl}\,(C) \setminus C$. Then there exists a sequence $\{x_k\}_k$ such that $x_k \to \bar{x}$ and $x_k \in C$ for all k. Apply then the previous argument to all x_k. Define $x_{k,\alpha} = \alpha x + (1-\alpha)x_k \implies B(x_{k,\alpha}, r\alpha) \cap \mathrm{aff}\,(C) \subseteq C$ for all k and $\alpha \in (0,1)$. Then the sequence $x_{k,\alpha} \to x_\alpha = \alpha x + (1-\alpha)\bar{x}$. For k large enough, $B\left(x_\alpha, \frac{\alpha r}{2}\right) \subseteq B(x_{k,\alpha}, r\alpha)$. Take the intersection with the affine hull, hence $B\left(x_\alpha, \frac{\alpha r}{2}\right) \cap \mathrm{aff}\,(C) \subseteq B(x_{k,\alpha}, r\alpha) \cap \mathrm{aff}\,(C)$, showing that $x_\alpha \in \mathrm{ri}\,(C)$.

■

Chapter 7

Lecture 7 of 24/03/2026

Proposition 1 (Nonemptyness of relative interiors). *Let $C \subseteq \mathbb{R}^d$ be convex and nonempty. Then:*

a) *the set* $\operatorname{ri}(C)$ *is a nonempty convex set,* $\operatorname{aff}(\operatorname{ri}(C)) = \operatorname{aff}(C)$*;*

b) *if* $m = \dim_{\mathbb{R}}(\operatorname{aff}(C))$, $m > 0$ *then there exists* $m+1$ *vectors* $x_0, x_1, \dots, x_m \in \operatorname{ri}(C)$ *such that*

$$\operatorname{aff}(C) = \operatorname{span}\{x_1 - x_0, x_2 - x_0, \dots, x_m - x_0\}.$$

Proof. The fact that $\operatorname{ri}(C)$ is convex is a consequence of LSP. We have to show that $\operatorname{ri}(C)$ is nonempty. Without loss of generality, we can assume $0 \in C$ (up to a traslation), consider $\operatorname{aff}(C)$. If $m = 0$ then $C = \{0\} \implies \operatorname{ri}(C) = C$ as for any $r > 0$, $B(0,r) \cap \operatorname{aff}(C) = C$. Assume now $m > 0$. Consider $z_1, \dots, z_m \in C$ linearly independent such that $\operatorname{aff}(C) = \operatorname{span}\{z_1, \dots, z_m\}$. Considet the set

$$X = \left\{ x : x = \sum_{i=1}^{m} \alpha_i z_i,\ \alpha_i > 0,\ \sum_i \alpha_i < 1 \right\}.$$

The strategy now is to prove that $X \subseteq \operatorname{ri}(C)$. Observe that $X \subseteq C$. Fix $\bar{x} \in X$, $x \in \operatorname{aff}(C)$ and observe that

$$\begin{aligned} \bar{x} &= Z\bar{\alpha},\ \bar{\alpha} \in \mathbb{R}^m \\ x &= Z\alpha\ \alpha \in \mathbb{R}^m \end{aligned}$$

with Z the matrix with coloumns $z_1, \dots, z_m$. Notice that

$$\begin{aligned} \|x - \bar{x}\|^2 &= \langle Z(\alpha - \bar{\alpha}), Z(\alpha - \bar{\alpha}) \rangle \\ &= (\alpha - \bar{\alpha})^\top Z^\top Z(\alpha - \bar{\alpha}) \\ &\leq \gamma \|\alpha - \bar{\alpha}\|^2. \end{aligned}$$

Since $\bar{x} \in X$, the vecotr $\bar{\alpha} \in A = \{\alpha \in \mathbb{R}^m : \sum_i \alpha_i < 1,\, \alpha_i > 0\}$ which is an open set. If x is suficently close to $\bar{x}$, then α is clost to $\bar{\alpha}$ by the above estimate on the norm. This implies that $\alpha \in A$, hence $x \in X$, so $B(\bar{x}, r) \cap \operatorname{aff}(C) \subseteq X \subseteq C$. This shows that $X \subseteq \operatorname{ri}(C)$.

Notice that, by construction, $\operatorname{aff}(X) = \operatorname{aff}(C)$, hence $\operatorname{aff}(X) \subseteq \operatorname{aff}(\operatorname{ri}(C)) \subseteq \operatorname{aff}(C)$.

b): fix $x_0 \in \operatorname{ri}(C)$ (x_0 exists by a)), translate C to $C - x_0$, so that $0 \in C - x_0$. fix $z_1, \ldots, z_m \in C - x_0$ linearly independent such that $\operatorname{aff}(C - x_0) = \operatorname{span}\{z_1, \ldots, z_m\}$. Let $\alpha \in (0,1)$ and consider $(1-\alpha) \cdot 0 + \alpha z_i \in \operatorname{ri}(C - x_0)$ by LSP as $0, z_i \in \operatorname{ri}(C - x_0)$. Hence $\alpha z_i \in \operatorname{ri}(C - x_0)$. It follows that the vectos $x_i = x_0 + \alpha z_i$ are such that $x_1 - x_0, \ldots, x_m - x_0$ spans the parallel space.

■

Lemma 2 (Prolungation lemma)**.** *Let $C \subseteq \mathbb{R}^d$ convex nonempty. A vector $x \in C$ belongs to* $\operatorname{ri}(C)$ *if and only if for any $\bar{x} \in C$ there exists a $\gamma > 0$ such that $x + \gamma(x - \bar{x}) \in C(*)$, i.e. every line segment connecting x to any $\bar{x}$ can be prolongated beyond x without leaving C.*

Proof. "$\Longrightarrow$" easy.

Let x satisfy $(*)$, fix $\bar{x} \in \operatorname{ri}(C)$ (it exists thanks to the above proposition). If $x = \bar{x}$ then $x \in \operatorname{ri}(C)$ and we are happy. If not, there exists a $\gamma > 0$ such that $z = x + \gamma(x - \bar{x}) \in C$ so that x lies within the line segment connecting $\bar{x}$ and z. By LSP, it follows that $x \in \operatorname{ri}(C)$.

■

Proposition 3 (Minima of concave funcitons)**.** *Let $X \subseteq \mathbb{R}^n$ be convex and nonempty. Let $f : X \to \mathbb{R}$ be concave. Define*

$$X^* = \arg\min_{x \in X} f(x).$$

If $X^* \cap \operatorname{ri}(X) \neq \emptyset$, then f is constant.

Proof. Assume $x^* \in X^* \cap \operatorname{ri}(X)$, by prolongation lemma, there exists $\gamma > 0$ such that the point $\hat{x} = x^* + \gamma(x^* - x)$ belongs to X, implying that

$$x^* = \frac{1}{\gamma+1}\hat{x} + \frac{\gamma}{\gamma+1}x.$$

By concavity of f,

$$
\begin{aligned}
f(x^*) &= f(\frac{1}{\gamma+1}\hat{x} + \frac{\gamma}{\gamma+1}x) \\
&\geq \frac{1}{\gamma+1}f(\hat{x}) + \frac{\gamma}{\gamma+1}f(x).
\end{aligned}
$$

Since x^* is a minimum point, it must be that $f(y) \geq f(x^*)$ for any y, in particular, $f(\hat{x}) \geq f(x^*)$ and $f(x) \geq f(x^*)$, hence $f(x) = f(x^*)$ for any x.

∎

An immediate consequence is that if $f(x) = a^\top x$ with $a \in \mathbb{R}^n$, then the minima of this function is attained at the boundary of its domain (which is assumed to be convex).

Proposition 4 (Continuity of convex funcitons). *If $f : \mathbb{R}^n \to \mathbb{R}$ is convex, then f is continous. More generally, if $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ is proper and convex, then the restriction of f to* $\mathrm{dom}\,(f)$ *is continous in* $\mathrm{ri}\,(\mathrm{dom}\,(f))$.

Proof. Assume $0 \in \mathrm{dom}\,(f)$ (if not translate). Without loss of generality, assume that the set $X = \{x : \|x\|_\infty \leq 1\} \subseteq \mathrm{dom}\,(f)$ (up to dilatation). It is now sufficent to prove that f is continous at 0: for any sequence $\{x_k\}_k$ such that $kx_k \to 0$ we want to prove $f(x_k) \to f(0)$. Define e_i, for $i = 1, \ldots, 2^n$ the corners of X. Let $x \in X$, then

$$
x = \sum_{i=1}^{2^n} \alpha_i e_i \quad \sum_i \alpha_i = 1,\, \alpha_i \geq 0.
$$

Then

$$
\begin{aligned}
f(x) &= f\left(\sum_{i=1}^{2^n} \alpha_i e_i\right) \\
&\leq \sum_i \alpha_i f(e_i) \\
&\leq \sum_i \alpha_i \left(\max_j f(e_j)\right) \\
&\leq \max_j f(e_j).
\end{aligned}
$$

Consider now the sequence $x_k \to 0$, define $y_k = x_k/\|x_k\|$, $z_k = -x_k/\|x_k\|$. We can see x_k as a convex combination of y_k and 0, hence

$$\begin{aligned} f(x_k) &= f(\|x_k\|_\infty y_k - (1 - \|x_k\|_\infty) \cdot 0) \\ &\leq \|x_k\| f(y_k) + (1 - \|x_k\|) f(0). \end{aligned}$$

Observe that $\|x_k\| \to 0$, and $f(y_k)$ is bounded ($y_k \in \text{dom}(f)$), hence $\|x_k\| f(y_k) \to 0$ as $k \to +\infty$. This shows that

$$\limsup_k f(x_k) \leq f(0).$$

Do similarly with z_k:

$$f(0) = f\left(\frac{1}{1 + \|x_k\| x_k} + \frac{\|x_k\|}{1 + \|x_k\| 0}\right) \leq \liminf_k f(x_k).$$

Hence $f(0) \leq \liminf_k f(x_k) \limsup_k f(x_k) \leq f(0)$. This shows that f is continous at 0.

■

7.1 Intermezzo

Let $E \subseteq \mathbb{R}^{n+1}$, $f : D \to \overline{\mathbb{R}}$ where $D = \{x \in \mathbb{R}^n : \exists\, w \in \mathbb{R} \text{ s.t } (x, w) \in E\}$, $f(x) = \inf_w (x, w) \in E$.

Let now $f : X \to \overline{\mathbb{R}}$ any function. We define the *closure of f* as

$$\text{cl}(f)(x) = \inf \{w : (x, w) \in \text{cl}(\text{epi}(f))\}.$$

The *convex closure* of f is

$$\check{\text{cl}}(f)(x) = \inf \{w : (x, w) \in \text{cl}(\text{conv}(\text{epi}(f)))\}.$$

It can be proven that $\check{\text{cl}}(f) = \text{cl}(F)$ where

$$F(x) = \inf \{w : (x, w) \in \text{conv}(\text{epi}(f))\}.$$

Proposition 1. *Let $f : X \to \overline{\mathbb{R}}$. Then*

$$\inf_{x \in X} f(x) = \inf_{x \in \mathbb{R}^n} \text{cl}(f)(x) = \inf_{x \in \mathbb{R}^n} F(x) = \inf_{x \in \mathbb{R}^n} \check{\text{cl}}(f)(x).$$

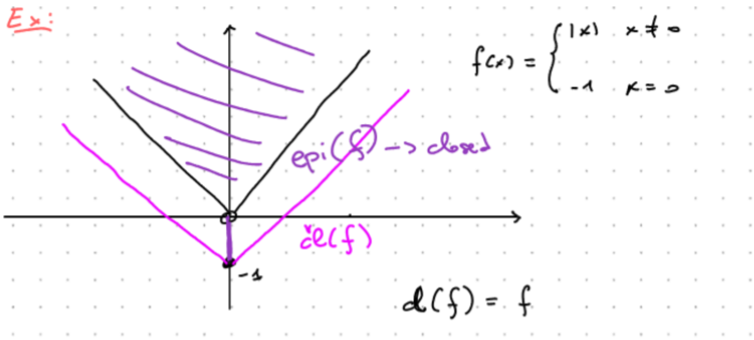


Figure 7.1: Example of convex closure of a function

Proof. If $\text{epi}\,(f) = \emptyset$, then $f \cong +\infty$, so this case is verified. Suppose now $\text{epi}\,(f) \neq \emptyset$. Let $f^* = \inf_{x\in\mathbb{R}^n} \text{cl}\,(f)\,(x)$. for any sequence $\{(\bar{x}_k, \bar{w}_k)\}_k \subseteq \text{cl}\,(\text{epi}\,(f))$ with $\bar{w}_k \to f^*$ we can construct a subsequence $\{(x_k, w_k)\}_k \subseteq \text{epi}\,(f)$ such that $w_k \to f^*$, $|w_k - \bar{w}_k| \to 0$. For each $x_k \in X$, it holds $(x_k, w_k) \in \text{epi}\,(f)$, then $f(x_k) \leq w_k$. But then

$$\inf_{x\in X} f(x) \leq \limsup_k f(x_k) \leq \lim_k w_k = f^* \leq \text{cl}\,(f)\,(x) \leq f(x).$$

Hence

$$\inf_{x\in X} f(x) = \inf_{x\in\mathbb{R}^n} \text{cl}\,(f)\,(x).$$

The next part uses the same argument.

■

Chapter 8

Lecture 8 of 25/03/2026

Proposition 1. *Let $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ a function. Then*

a) $\mathrm{cl}\,(f)$ *is the greatest closed function below f (recall that we say $g \leq f$ if $g(x) \leq f(x)$ for any x);*

b) $\check{\mathrm{cl}}\,(f)$ *is the greatest convex closed function below f.*

Proof. Let $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ closed below f, then $\mathrm{epi}\,(f) \subseteq \mathrm{epi}\,(g)$, but $\mathrm{epi}\,(\mathrm{cl}\,(f)) = \mathrm{cl}\,(\mathrm{epi}\,(f)) =$ intersection of all closed sets containing $\mathrm{epi}\,(f)$. Notice that $\mathrm{epi}\,(g)$ is closed, as g is closed. Hence $\mathrm{cl}\,(\mathrm{epi}\,(f)) \subseteq \mathrm{epi}\,(g)$ as $\mathrm{epi}\,(g)$ is one of the closed sets containing $\mathrm{epi}\,(f)$. This shows that $\mathrm{cl}\,(f) \leq g$. The same argument works for b).

■

8.1 Recession cones

Definition 1 (Recession direction/cone)**.** Let C be convex and nonempty. We say that a direction $d \in \mathbb{R}^n$ is a *recession direction* for C if $x + \alpha d \in C$ for any $x \in C$, $\alpha \geq 0$.

We denote with R_C the *recession cone* of C the set of all recession directions.

Example 2.

$$C = \left\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\right\}. \quad \mathrm{R}_C = \{0\}. \tag{8.1}$$

$$C = (-\infty, 2]. \quad \mathrm{R}_C = (-\infty, 0]. \tag{8.2}$$

Proposition 3 (Recession cone theorem)**.** *Let C closed convex nonempty. Then d functions. This methodology is fundamental in several convex optimization contexts, including the issue of existence of optimal solutions, which will be discu*

a) R_C *is closed and convex;*

b) $d \in \mathrm{R}_C \iff$ *there exists $x \in C$ such that $x + \alpha d \in C$ for all $\alpha \geq 0$.*

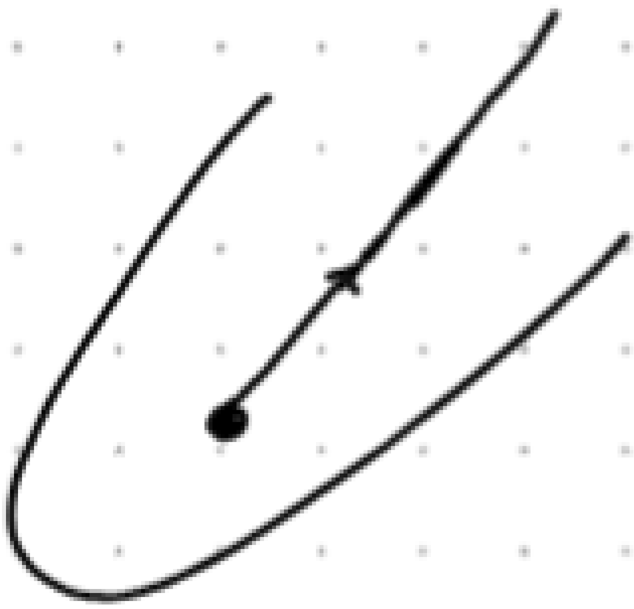


Figure 8.1: Example of recession cone

Proof. We first prove convexity. Let $d_1, d_2 \in \mathrm{R}_C$, $\gamma_1, \gamma_2 \in [0,1]$ ssuch that $\gamma_1 + \gamma 2 = 1$. Fix any $x \in C$, $\alpha \geq 0$ and consider $x + \alpha(\gamma_1 d_1 + \gamma_2 d_2) = x(\gamma_1 + \gamma_2) + \alpha(\gamma_1 d_1 + \gamma_2 d_2) = \gamma_1(x + \alpha d_1) + \gamma_2(x + \alpha d_2)$ which is a convex combination of points in C. For the closerness, consider $\{d_k\}_k$ a sequence in R_C such that $d_k \to d \in \mathbb{R}^n$. Then for any $x \in C$ and for any $\alpha \geq 0$, $x + \alpha d_k \in C \to x + \alpha d$ as $k \to +\infty$. This shows that $x + \alpha d \in C$ as C is closed.

For point b), the implication $'' \implies ''$ is obvious.

Let $d \in \mathbb{R}^n$ and assume $x \in C$ such that $x + \alpha d \in C$ for any $\alpha \geq 0$. Let $\bar{x} \in C$. We have to prove $\bar{x} + \alpha d \in C$. Assume, without loss of generality, $\alpha = 1$ as we can retrive the general case with a scaling argument. Define a sequence $z_k = x + kd$, $z_k \in C$ for all k. If $\bar{x} = z_k$ for some k then we are done. Assume then $\bar{x} \neq x_k$ and define $d_k \dfrac{z_k - \bar{x}}{\|z_k - \bar{x}\|}\|d\|$. Then

$$\begin{aligned} d_k &= \frac{z_k - \bar{x} + x - x}{\|z_k - \bar{x}\|}\|d\| = \frac{\overbrace{z_k - x}^{=kd}}{\|z_k - \bar{x}\|}\|d\| + \frac{x - \bar{x}}{\|z_k - \bar{x}\|} \\ &= \frac{k\|d\|}{\|z_k - \bar{x}\|}d + fracx - \bar{x}\|z_k - \bar{x}\|. \end{aligned}$$

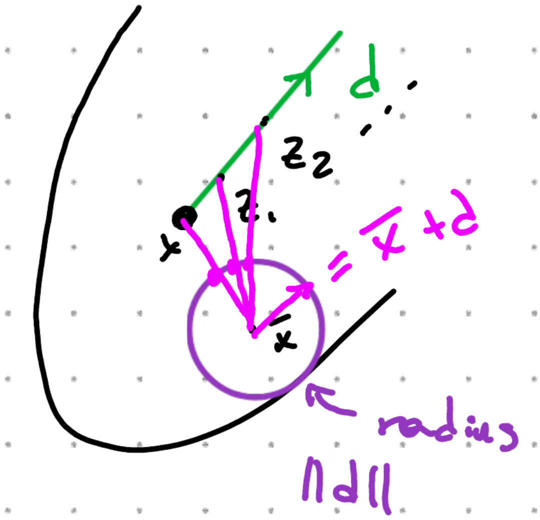


Figure 8.2: Visual intuition on the sequence z_k

The last term goes to 0, indeed, $x - \bar{x}$ is fixed, $\|d\|$ is fixed and $\|z_k\| \nearrow +\infty$ as $k \nearrow +\infty$. I claim that the first term goes to d: in fact,

$$\frac{k\|d\|}{\|x + kd - \bar{x}\|} = \frac{k\|d\|}{k\|d + (x - \bar{x})/k\|} \to 1.$$

This shows that $d_k \to d$ as $k \nearrow +\infty$. For k large enough, the vector $\bar{x} + dk$ belongs to the segment connecting $\bar{x}$ to z_k, so, by convexity of C, $\bar{x} + d_k \in C$, hence $\bar{x} + d \in C$ as C is closed.

■

Proposition 4 (Properties of the recession cone). *Let $C \subseteq \mathbb{R}^n$ closed convex nonempty. Then*

a) R_C *contains a nonzero direction if and only if C is unbounded;*

b) $\mathrm{R}_C = \mathrm{R}_{\mathrm{ri}(C)}$

c) if $\{C_i\}_{i \in I}$ *is a collection of closed convex sets such that* $\bigcap_{i \in I} C_i \neq \emptyset$*, then* $\mathrm{R}_{\bigcap_i C_i} = \bigcap_{i \in I} \mathrm{R}_{C_i}$*;*

d) fix $W \subseteq \mathbb{R}^n$ *compact,* $A \in \mathbb{R}^{n \times n}$ *a matrix. Let* $V = \{x \in C : Ax \in W\}$*. Assume V nonempty. Then* $\mathrm{R}_V = \mathrm{R}_C \cap N(A)$ *where* $N(A) = \{x \in \mathbb{R}^n : Ax = 0\}$ *is the null space of the linear operator A.*

Proof. Not required. ■

Example 5. The following example shows how the closerness of C can play a role in the recession cone. Consider the set

$$C = \left\{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1,\, y \geq 0\right\} \cup \{(1, 0)\}.$$

Then $\mathrm{R}_C = \{0\}$, indeed, if we start from the point $(1, 0)$, we cannot move along any direction without leaving C, while $\mathrm{R}_{\mathrm{ri}(C)} = \operatorname{span}\{(0, 1)\}$.

8.2 Linearity space

We define the linearity space for a set C as the set

$$\mathrm{L}_C = \mathrm{R}_C \cap (-\mathrm{R}_C).$$

In other words, $d \in \mathrm{L}_C$ if and only if $x + \alpha d \in C$ and $x - \alpha d$ are inside C for any $\alpha \geq 0$.

Proposition 1 (Properties of L_C). *Let $C \subseteq \mathbb{R}^n$ convex closed nonempty, then*

1. L_C *is a subspace;*
2. $\mathrm{L}_C = \mathrm{L}_{\mathrm{ri}(C)}$*;*
3. *same as above.*

Proof. Not required. ■

Example 2 (Important!). Consider the set

$$C = \left\{x \in \mathbb{R}^n : x^\top Q x + c^\top x + b \leq 0\right\}$$

where $Q \in \mathbb{R}^{n \times n}$ is SPD, $c \in \mathbb{R}^n$, $b \in \mathbb{R}$. We want to compute R_C.
Let $x \in C$, $d \in \mathbb{R}^n$. Then $x + \alpha d \in C$ for any $\alpha \geq 0$ if and only if

$$\begin{aligned} &(x + \alpha d)^\top Q (x + \alpha d) + c^\top (x + \alpha d) + b \leq 0 \\ \iff\ &\underbrace{x^\top Q x + c^\top x + b}_{\leq 0} + \alpha (c^\top d + 2x^\top Q) d + \alpha^2 d^\top Q d \leq 0 \end{aligned}$$

for any $\alpha \geq 0$ and for any $x \in C$.
The quantity $\alpha^2 d^\top Q d \geq 0$ since Q is SPD, hence we need $d^\top Q d = 0$. Since Q is SPD, we can compute its Cholesky factorization, hence $Q = M^\top M \implies d^\top M^\top M d = 0$, so $Md = 0$.

Furthermore, we need also $\alpha c^\top d \leq 0$. Finally, the recession cone turns out to be

$$\mathrm{R}_C = \left\{ d \in \mathbb{R}^n : Qd = 0,\, c^\top d \leq 0 \right\}.$$

Moreover,

$$\mathrm{L}_C = \left\{ d \in \mathbb{R}^n : Qd = 0,\, c^\top d = 0 \right\}.$$

Chapter 9

Lecture 9 of 31/03/2026

Proposition 1. *Let $C \subseteq \mathbb{R}^n$ nonempty convex, let $S \subseteq \mathrm{L}_C$ be a subspace. Then we can decompose C as*

$$C = S + (C \cap S^{\perp}).$$

Proof. Not required. ■

Proposition 2. *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be closed proper convex funciton. Let $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ for all $\gamma \in \mathbb{R}$ (sublevel sets of f). Then*

a) all nonempty V_γ have the same recesison cone, denoted by

$$\mathrm{R}_f = \left\{ d \in \mathbb{R}^n : (d, 0) \in \mathrm{R}_{\mathrm{epi}(f)} \right\}$$

b) if one V_γ is compact, then all the sublevel sets are compact.

Proof. Fix $\gamma \in \mathbb{R}$, assume $V_\gamma \neq$. Define the γ-slice $S = \{(x, \gamma) : f(x) \leq \gamma\}$. Observe that $S = \mathrm{epi}\,(f) \cap \{(x, \gamma) : x \in \mathbb{R}^n\}$. We already know $\mathrm{epi}\,(f)$ closed convex, so

$$\mathrm{R}_S = \mathrm{R}_{\mathrm{epi}(f)} \cap \mathrm{R}_{\{(x,\gamma):x\in\mathbb{R}^n\}} = \mathrm{R}_{\mathrm{epi}(f)} \cap \{(d, 0) : d \in \mathbb{R}^n\} = \left\{ (d, 0) : d \in \mathrm{R}_{\mathrm{epi}(f)} \right\}$$

which does not depend on γ.

Point b): form a), we know $\mathrm{R}_{V_{\gamma_1}} = \mathrm{R}_{V_{\gamma_2}}$ for any $\gamma_1, \gamma_2 \in \mathbb{R}$ such that their sublevel sets are nonempty. If, for $\bar{\gamma} \in \mathbb{R}$, $V_{\bar{\gamma}}$ is compact, then $\mathrm{R}_{V_{\bar{\gamma}}} = \{0\}$. Then for a) every recession cone is equal, hence $\mathrm{R}_{V_\gamma} = \{0\}$ for any γ. This shows that V_γ is compact for every $\gamma \in \mathbb{R}$.

■

Remark 3. Let $d \in \mathrm{R}_f$ if $d \in \mathrm{R}_{V_{\bar{\gamma}}}$ for any γ. Fix $\gamma \in \mathbb{R}$, then $d \in \mathrm{R}_{V_\gamma}$ implies that for $x \in V_\gamma$, $x + \alpha d \in V_\gamma$ for any $\alpha \geq 0$, hence $f(x + \alpha d) \leq \gamma$ for every $\alpha \geq 0$. This shows that on the direction d, the function f cannot increase, so d is a direction of continous non-ascent for f.

Definition 4 (Linearity space of a function)**.** The linearity space of a function is defined in the same way as the linearity space of a set.

Remark 5. If $d \in \mathrm{L}_f$, then

$$f(x + \alpha d) \leq f(x) \quad \forall \alpha \geq 0$$
$$f(x - \alpha d) \leq f(x) \quad \forall \alpha \geq 0.$$

I claim that d is a constancy direction for f. Let $x = \frac{1}{2}(x - \alpha d) + \frac{1}{2}(x + \alpha d)$. Indeed, assume by contraddiction $f(x - \alpha d) < f(x)$. Then

$$\begin{aligned} f(x) &= f\left(\frac{1}{2}(x - \alpha d) + \frac{1}{2}(x + \alpha d)\right) \\ &\leq \frac{1}{2}\underbrace{f(x - \alpha d)}_{< f(x)} + \frac{1}{2}\underbrace{f(x + \alpha d)}_{\leq f(x)} < f(x). \end{aligned}$$

which is a contraddiction.

This shows that $f(x + \alpha d) = f(x - \alpha d) = f(x)$ for any $\alpha \geq 0$. The space L_f is also called *constancy space*.

Example 6. Consider $f(x) = x^\top Q x + c^\top x + b$ as before. It is not difficoult to show that

$$\mathrm{L}_f = \left\{ d : Qd = 0,\, c^\top d = 0 \right\}.$$

9.1 Recession function

Consider $f : \mathbb{R}^n \to (-\infty, +\infty]$ closed proper convex. This implies $\operatorname{epi}(f)$ is closed convex subset of $\mathbb{R}^{n+1}$. Consider $\mathrm{R}_{\operatorname{epi}(f)} \subseteq \mathbb{R}^{n+1}$. Question: is that set the epigraph of something? The answer is yes, for every x, the quantity $\inf_w \left\{ (d, w) : d \in \mathrm{R}_{\operatorname{epi}(f)} \right\}$ is finite or $+\infty$. Since $(d, w) \in \mathrm{R}_{\operatorname{epi}(f)}$ then $(d, f(x)) + \alpha(d, w) \in \operatorname{epi}(f)$, hence $(x + \alpha d, f(x) + \alpha w) \in \operatorname{epi}(f) \implies f(x + \alpha d) \leq f(x) + \alpha w$. Assume w can be taken as negative as we want, so $f(x + d) = -\infty$, but this is not possible. This implies that $\mathrm{R}_{\operatorname{epi}(f)}$ can be the epigraph of some function.

Definition 1 (Recession function)**.** We define r_f the recession function of f as the function such that $\operatorname{epi}\left(\mathrm{r}_f\right) = \mathrm{R}_{\operatorname{epi}(f)}$.

Chapter 10

Lecture 10 of 01/04/2026

Proposition 1 (Book 1.4.6). *Let $f : \mathbb{R} \to (-\infty, +\infty]$ closed proper convex, Consider R_f and L_f the recession cone and constancy space. Then*

$$\begin{aligned} \mathrm{R}_f &= \left\{ d \in \mathbb{R}^n : \mathrm{r}_f(d) \leq 0 \right\} \\ \mathrm{L}_f &= \left\{ d \in \mathbb{R}^n : \mathrm{r}_f(d) = 0 \right\} \end{aligned}$$

Proof. It follows from the definition of recession cone of a function that

$$\begin{aligned} \mathrm{R}_f &= \left\{ d \in \mathbb{R}^n : (d, 0) \in \mathrm{R}_{\mathrm{epi}(f)} \right\} \\ &= \left\{ d \in \mathbb{R}^n : (d, 0) \in \mathrm{epi}\left(\mathrm{r}_f\right) \right\} \\ &= \left\{ d \in \mathbb{R}^n : \mathrm{r}_f \leq 0 \right\} \end{aligned}$$

For the linearity space instead, we observe that $\mathrm{L}_f = \mathrm{R}_f \cap (-\mathrm{R}_f)$, $\mathrm{r}_f(d) \leq 0$ and $\mathrm{r}_f(-d) \leq 0$ for any $d \in \mathrm{L}_f$. Then

$$\mathrm{r}_f(0) = \mathrm{r}_f\left(\frac{1}{2}d + \frac{1}{2}(-d)\right) \leq \frac{1}{2}\mathrm{r}_f(d) + \frac{1}{2}\mathrm{r}_f(-d).$$

Observe that $\mathrm{r}_f(0) = 0$. Indeed, $\mathrm{r}_f(0) = \inf\left\{ w : (0, w) \in \mathrm{R}_{\mathrm{epi}(f)} \right\} = 0$ because we can take w as negative as we want, but f is proper, hence that infimum id 0. It follows that $\mathrm{r}_f(d) = \mathrm{r}_f(-d) = 0$ for any $d \in \mathrm{L}_f$.

■

Proposition 2 (Book 1.4.7). *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be closed proper convex. Then, for every $x \in \mathrm{dom}\,(f)$, for any $d \in \mathbb{R}^n$ we have*

$$\mathrm{r}_f(d) = \sup_{\alpha > 0} \frac{f(x + \alpha d) - f(x)}{\alpha} = \lim_{\alpha \to +\infty} \frac{f(x + \alpha d) - f(x)}{\alpha}.$$

Intuitively it gives a measure on "how fast f is diverging in the direction d".

Proof. Not required.

■

10.1 Nonemptyness of instersection of closed sets

Now we ask ourselfs a question: for $\{C_k\}_k$ a sequence of sets in $\mathbb{R}^n$ nested, i.e. $C_{k+1} \subseteq C_k$ for any k. We know that C_k is compact for any k, then $\bigcap_k C_k \neq \emptyset$. Can we do similarly also if C_k are only closed?

Why we ask such a question?

Example 1. Assume that we are looking for $\inf_{x\in X} f(x)$. Consider a sequence $\{\gamma_k\}_k \subseteq \mathbb{R}^n$, $\gamma_k \to \gamma^* = \inf_{x\in X} f(x)$. Does such a point exist? Define $X_k = \{x : f(x) \leq \gamma_k\}$. One can show that if $\bigcap\limits_k X_k$ is nonempty then there exists x^* such that $f(x^*) = \gamma^*$.

Example 2. Consider $A : \mathbb{R}^n \to \mathbb{R}^n$ a linear map, C a closed set. Is the set AC closed? Not always. Indeed, fix a sequence $\{y_k\}_k \subseteq AC$, $y_k \to \bar{y} \in \mathbb{R}^n$. Does $\bar{y} \in C$? Set $C_k = C \cap N_k$ where $N_k = \{x : \|Ax - \bar{y}\| \leq \|y_k - \bar{y}\|\}$. One can show that if the intersection of C_k is nonempty, then there exists a point $\bar{x} \in \bigcap_k C_k$ such that $A\bar{x} = \bar{y}$.

From now on, C_k denotes a sequence of nested nonempty convex closed sets.

Definition 3. We say that a sequence $\{x_k\}_k$, $x_k \neq 0$ is an *asymptotic sequence* of $\{C_k\}_k$ if $\|x_k\| \to +\infty$ and

$$\frac{x_k}{\|x_k\|} \to \frac{d}{\|d\|}$$

where $d \neq 0$, $d \in \bigcap_k \mathrm{R}_{C_k}$. Intuitively, we can say that $\{x_k\}_k$ "explodes only in a particular direction".

Example 4. Consider $C_k = \{x : x \geq k\}$, then $d = 1 \in \bigcap_k \mathrm{R}_{C_k}$. Moreover, any unbounded sequence in $\mathbb{R}$ is asymptotic, as $\mathbb{R}$ has a unique direction in which a sequence can escape.

Remark 5. Given any unbounded sequence $\{x_n\}_n \subseteq \mathbb{R}^d$, there exists a subsequence $\{x_{n_k}\}_k$ that is asymptotic, because every cluster point of $\{x_n/\|x_n\|\}_n$ is a common recession direction for each C_k.

Definition 6 (Retractive asymptotic sequence)**.** An asymptotic sequence $\{x_k\}_k$ of $\{C_k\}_k$, with d as asymptotic direction is *retractive* if there exists a $\bar{k}$ big enough such that $x_k - d \in C_k$ for any $k > bark$.

Definition 7. A sequence $\{C_k\}_k$ is *retractive* if all its asymptotic sequence are retractive. A set $C \subseteq \mathbb{R}^n$ is retractive in the same way.

Example 8. The sequence $C_k = \{x : x \geq k\}$ is not retractive because there exist a sequence $\{x_k\}_k$ such that $x_k - d \notin C_k$, e.g. $x_k = k, d = 1$, so $x_k - d = k - 1 \notin C_k$.

Let $C_k = \left\{(x, y) \in \mathbb{R}^2 : |x| \leq \frac{1}{k}\right\}$. Then

$$\bigcap_k C_k = \{(0, y) : y \in \mathbb{R}\} \neq \emptyset.$$

This sequence is retractive, as the only direction of escape is y.

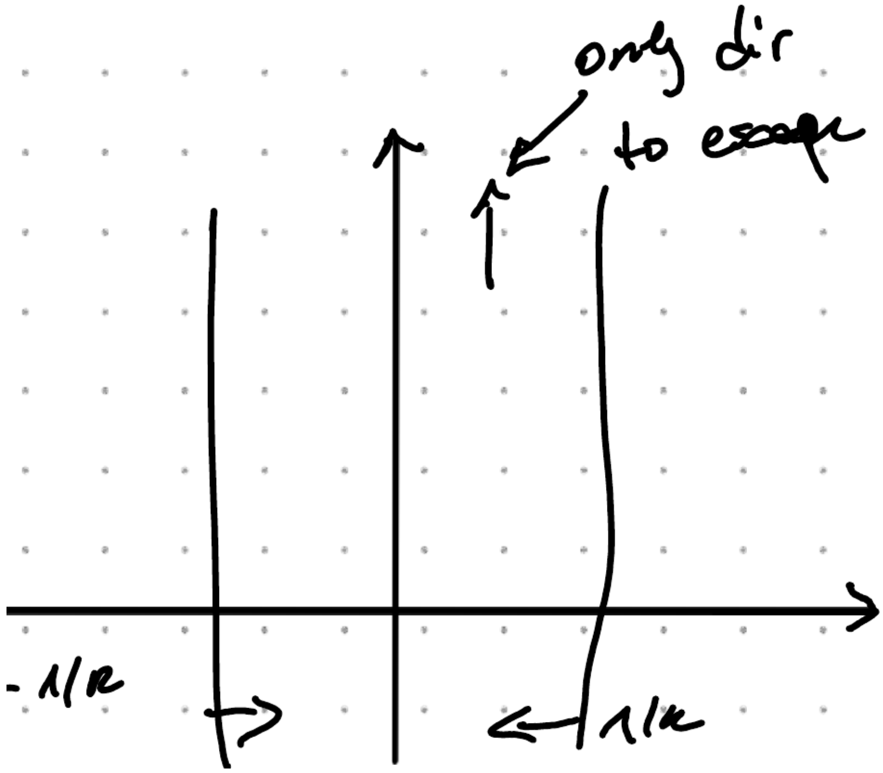


Figure 10.1: Visual intuition for the example

Remark 9. Assume $\left\{C_k^1\right\}_k, \left\{C_k^2\right\}_k, \ldots, \left\{C_k^m\right\}_k$ to be m families of nested closed convex nonempty sets. Then define

$$N_k = C_k^1 \cap \cdots \cap C_k^m$$
$$T_k = C_k^1 \times \cdots \times C_k^m.$$

If each sequence is retractive and $N_k \neq \emptyset$ then $\{N_k\}_k$ and $\{T_k\}_k$ are retractive.

Proposition 10. *Polyhedral sets are retractive.*

Proof. Corollary of the above remark. ∎

Proposition 11. *A retractive nested sequence of nonempty closed convex sets has nonempty interior.*

Proof. Scketched in class, not reported here. ∎

Example 12. Retractiveness is *not* a necessary condition: consider the sequence

$$C_k = \left\{ (x,y) \in \mathbb{R}^2 : y \geq x^2 - \frac{1}{k} \right\}.$$

Then the intersection is nonempty but C_k is not retractive: the only direction of escape is $(0,1)$, but if we take a point on the boundary of each C_k we go out of the set.

The question now is: how can we easily prove retractiveness? The answer is in the following:

Proposition 13 (Book 1.4.11)**.** *Denote $R = \bigcap_k \mathrm{R}_{C_k}$ and $L = \bigcap \mathrm{L}_{C_k}$. Then*

a) Il $R = L$ then $\{C_k\}_k$ is retractive. Furthermore it holds true that

$$\bigcap_k C_k = L + \tilde{C}$$

where $\tilde{C}$ is a nonempty compact set;

b) Fix $X \subseteq \mathbb{R}^n$ retractive, closed and convex. Assume that the sets $\overline{C}_k = X \cap C_k$ are nonempty and $\mathrm{R}_X \cap R \subset L$. Then $\overline{C}_k$ is retractive.

Proof. Not required. ∎

Proposition 14 (Existance of solution of convex quadratic problems, book 1.4.12)**.** *Let $Q \in \mathbb{R}^{n \times n}$ be SPD, $c \in \mathbb{R}^n$, $a_1, \ldots, a_m \in \mathbb{R}^n$, $b_1, \ldots, b_m \in \mathbb{R}$. Let $f : \mathbb{R}^n \to \mathbb{R}$, $x \mapsto x^\top Q x + c^\top x$, and $X = \left\{ x \in \mathbb{R}^n : a_j^\top x \leq b_j, \forall 1 \leq j \leq m \right\}$. Assume $\inf_{x \in X} f(x)$ to be finite. Then there exists $x^* \in X$ such that*

$$f(x^*) = \inf_{x \in X} f(x).$$

Proof. Let $f^* = \inf_{x \in X} f(x)$. Let $\{\gamma_k\}_k \subseteq \mathbb{R}$ such that $\gamma_k \to f^*$ as $k \to +\infty$. Define $\overline{C}_k = X \cap \{x : f(x) \leq \gamma_k\}$. Let R_X be the recession cone of X, $R = \mathrm{R}_{\bigcap_k C_k}$, $L = \mathrm{L}_{\bigcap_k C_k} (= \mathrm{R}_{C_k} = \mathrm{L}_{C_k})$ since we already know that the sublevel sets share all the same recession cone. In a previous example we computed

$$
\begin{aligned}
R &= \left\{ d : Qd = 0,\, c^\top d \le 0 \right\} \\
L &= \left\{ d : Qd = 0,\, c^\top d = 0 \right\} \\
\mathrm{R}_X &= \left\{ d : a_j^\top d \le 0,\, \forall\, 1 \le j \le m \right\}.
\end{aligned}
$$

Take $d \in \mathrm{R}_X \cap R$, assume $d \notin L$. Then $Qd = 0$, $c^\top d < 0$, $a_j^\top d \le 0$ for all j, but then $f(x + \alpha d) \to -\infty$ as $\alpha \to +\infty$, which contraddicts the fact that f^* is finite. Hence $\mathrm{R}_X \cap R \subseteq L$, so by the previous theorem, $\left\{ \overline{C}_k \right\}_k$ is retractive, and so there exists $x^* \in \bigcap_k \overline{C}_k$ such that $f(x^*) = f^*$.

■

Chapter 11

Lecture 11 of 07/04/2026

Proposition 1 (Book 1.4.13). *Let $X, C \subseteq \mathbb{R}^n$ convex closed nonempty. Let $A \in \mathbb{R}^{m \times n}$, $m \in \mathbb{N}$, $m > 0$ and denote with $N(A)$ the nullspace of A. If X is retractive and $\mathrm{R}_X \cap \mathrm{R}_C \cap N(A) \subseteq \mathrm{L}_C$ then $A(X \cap C)$ is closed.*

Proof. Let $\{y_k\}_k \subseteq A(X \cap C)$ such that $y_k \to \bar{y}$ as $k \to +\infty$. To prove that $\bar{y} \in A(X \cap C)$, hence that there exists $\bar{x} \in X \cap C$ such that $A\bar{x} = \bar{y}$. Define $C_k = C \cap N_k$ where $N_k = \{x : \|Ax - \bar{y}\| \leq \|y_k - \bar{y}\|\}$, these C_K are convex and closed. Moreover $\mathrm{R}_{C_k} = \mathrm{R}_C \cap \mathrm{R}_{N_k} = \mathrm{R}_C \cap N(A)$. Use then point b) of proposition of the last lecture, and we are done. Why? $\|y_k - \bar{y}\| \to 0$. For every $x \in C_k$, $\|Ax - \bar{y}\| \leq \|y_k - \bar{y}\| \to 0$. For $x \in \bigcap_k C_k$, $\|Ax - \bar{y}\| \leq$ any value of a sequence that is going to 0, hence $\|Ax - \bar{y}\| = 0 \implies Ax = \bar{y}$. ■

Remark 2. If $C = \mathbb{R}^n$, then $\mathrm{L}_C = \mathbb{R}^n$, X a polyhedral set (hence retractive), then $A(C \cap X) = A \times X$ is closed.

Remark 3. $X = \mathbb{R}^n$, $\mathrm{R}_X = \mathbb{R}^n$. $A(X\ capC) = AC$ closed if $\mathrm{R}_{\cap} N(A) = \{0\}$.

Proposition 4. *Let $C_1, \dots C_m$ be nonempty closed convex sets such that, if $d_1 + \cdots + d_m = 0$ for $d_i \in \mathrm{R}_{C_i}$ then $d_i \in \mathrm{L}_{C_i}$, then $C_1 + \cdots + C_m$ is closed.*

Proof. Not required. ■

Remark 5. If C_1 and $-C_2$ are closed and convex, then $C_1 - C_2$ is closed convex if $\mathrm{R}_{C_1} \cap \mathrm{R}_{C_2} = \{0\}$, so if one betweeen C_1 and C_2 is bounded.

Example 6. Let

$$
\begin{aligned}
C_1 &= \{(x, y) : x \geq 0,\ y \geq 1/x\} \\
C_2 &= \{(x, y) : x = 0\}.
\end{aligned}
$$

Then

$$\begin{aligned} C_1 + C_2 &= \{(x_1, y_1) + (x_2, y_2) : (x_1, y_1) \in C_1, \, (x_2, y_2) \in C_2\} \\ &= \{(x_1, y_1 + \alpha) : (x_1, y_1) \in C_1, \, \alpha \in \mathbb{R}\} \\ &= \{(x, y) : x > 0\} \end{aligned}$$

which is an open set. Observe that

$$\begin{aligned} d_1 &= (0, 1) \in \mathrm{R}_{C_1} \\ d_2 &= (0, -1) \in \mathrm{R}_{C_2}. \end{aligned}$$

Then $d_1 + d_2 = 0$ and $d_2 \in \mathrm{L}_{C_2}$ but $d_1 \notin \mathrm{L}_{C_1}$.

11.1 Hyperplanes

An hyperplane is a set

$$H = \left\{ x : a^\top x = b \right\}$$

for some $a \in \mathbb{R}^n$, $a \neq 0$, $b \in \mathbb{R}$. For $\bar{x} \in H$, we have

$$H = \left\{ x : a^\top x = a^\top \bar{x} \right\}$$

hence $H = \bar{x} + \left\{ x : a^\top x = 0 \right\}$, where the latter is the parallel subspace to H. Furthermore we have the closed halfspaces where we replace the equal sign with a large inequality, closed with strict.

11.1.1 Separation

We say that H separates C_1 and C_2 if $a^\top x_1 \leq b \leq a^\top x_2$ or the converse. We also write

$$\sup_{x_1 \in C_1} a^\top x_1 \leq \inf_{x_2 \in C_2} a^\top x_2.$$

Assume $\bar{x} \in \operatorname{cl}(C)$ for $C \subseteq \mathbb{R}^n$. If H separates $\{\bar{x}\}$ from C, we say H is a supporting hyperplane at $\bar{x}$.

Proposition 1 (Supporting hyperplane theorem, book 1.5.1). *Let $C \subseteq \mathbb{R}^n$ nonempty convex, $\bar{x} \in \mathbb{R}^n$, $x \notin \operatorname{int}(C)$. Then there exists an hyperplane H that passes through $\bar{x}$ and contains C in one of its closed halfspaces, i.e. there exists $a \in \mathbb{R}^n$, $a \neq 0$ such that*

$$a^\top \bar{x} \leq a^\top x \quad \forall x \in C.$$

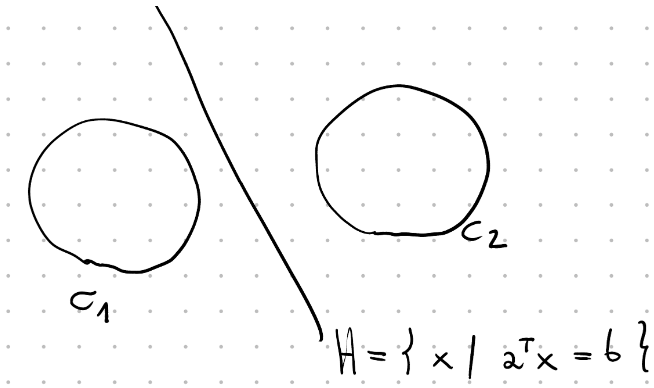


Figure 11.1: Hyperplane separating two closed convex sets

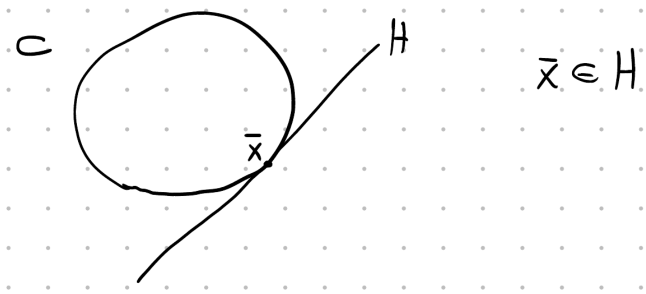


Figure 11.2: Example of supporting hyperplane

Proof. Consider $\bar{x} \notin \operatorname{int}(C)$. One can prove that there exists a sequence $\{x_k\}_k \subseteq \mathbb{R}^n \setminus \operatorname{cl}(C)$ such that $x_k \to \bar{x}$. (to prove this, later...)

Consider $\hat{x}_k$ the projection of x_k onto $\operatorname{cl}(C)$ which is a closed convex set, so the projection exists and it is uniqe. By projection theorem, it holds that

$$\langle \hat{x}_k - x_k, x - \hat{x}_k \rangle \geq 0 \quad \forall x \in \operatorname{cl}(C)\,.$$

Rewrite

$$\begin{aligned}
(\hat{x}_k - x_k)^\top \cdot x &\geq (\hat{x}_k - x_k)^\top \cdot (\hat{x}_k \pm x_k) \\
&= \underbrace{(\hat{x}_k - x_k)^\top \cdot (\hat{x}_k - x_k)}_{\geq 0} + (\hat{x}_k - x_k)^\top \cdot x_k \\
&\geq (\hat{x}_k - x_k)^\top \cdot x_k
\end{aligned}$$

Define $a_k = \dfrac{\hat{x}_k - x_k}{\|\hat{x}_k - x_k\|}$, we obtain that $a_k^\top \cdot x \geq a_k^\top \cdot x_k$ for any $x \in \operatorname{cl}(C)$

and for any k. The sequence $\{a_k\}_k \subseteq \overline{B(0,1)}$, so by BW, up to subsequences, $a_k \to \bar{a} \in \mathbb{R}^n$, $\|\bar{a}\| = 1$. Taking the limit

$$\begin{array}{ccc} a_k^\top \cdot x & \geq & \top_k x_k \\ \downarrow & & \downarrow \\ \bar{a}^\top \cdot x & \geq & \bar{a}^\top \cdot \bar{x}. \end{array}$$

Hence $\bar{a}$ defines our target hyperplane.

■

Proposition 2 (Separating hyperplane theorem (finite HB), Book 1.5.2). *Let $C_1, C_2 \subseteq \mathbb{R}^n$ be nonempty convex. If C_1 and C_2 are disjoint, then there exists $a \in \mathbb{R}^n$, $a \neq 0$ such that*

$$a^\top x_1 \leq a^\top x_2 \quad \forall x_1 \in C_1 \, \forall x_2 \in C_2.$$

Proof. Let $C = C_1 - C_2$ the algebraic set difference. Observe that $0 \notin C$, as $C_1 \cap C_2 = \emptyset$. Let H be the hyperplane separating $\{0\}$ and C, so there exists $a \in \mathbb{R}^n \setminus \{0\}$ such that $0 \leq a^\top x$ for all $x \in C$. Every $x \in C$ is in the form $x = x_1 - x_2$, hence

$$0 \leq a^\top (x_1 - x_2) \iff a^\top x_2 \leq a^\top x_1.$$

■

Example 3. Let

$$\begin{aligned} C_1 &= \{(x,y) : y \geq e^x\} \\ C_2 &= \{(x,y) : y \leq 0\}. \end{aligned}$$

Then $C_1 \cap C_2 = \emptyset$ and $H = \{(x,y) : y = 0\}$.

Proposition 4 (Strict separation theorem, book 1.5.3). *Let $C_1, C_2 \subseteq \mathbb{R}^n$ convex disjoint. Then C_1 and C_2 can be strictly separated if one of the following holds:*

1. *$C_1 - C_2$ is closed;*
2. *C_1 closed and C_2 compact;*
3. *C_1, C_2 polyhedral;*
4. *C_1, C_2 closed and $\mathrm{R}_{C_1} \cap \mathrm{R}_{C_2} = \mathrm{L}_{C_1} \cap \mathrm{L}_{C_2}$;*
5. *C_1 is closed, C_2 polyhedral, $\mathrm{R}_{C_1} \cap \mathrm{R}_{C_2} \subseteq \mathrm{L}_{C_1}$.*

Proof. We only prove under condition 2): assume $C_1 = A$ closed and $C_2 = B$ compact. Define

$$d^* = \inf_{a\in A,\, b\in B} \|a-b\| \geq 0.$$

Consider a maximizing sequence (a_k, b_k) where $a_k \in A, b_k \in B$ such that $\|a_k - b_k\| \to d^*$. Since B is compact, $\{b_k\}_k$ is bounded, hence, up to subsequences, $b_k \to \bar{b} \in B$, since B is closed. This implies also that $\{a_k\}_k$ is bounded (otherwise $\|a_k - b_k\| \to +\infty$), so, up to subsequences, $a_k \to \bar{a} \in A$. We have shown that $\|a_k - b_k\| \to \|\bar{a} - \bar{b}\|$. But, observing that $\|a_k - b_k\| \to d^*$, by uniqueness of the limit, we have also that $d^* = \|\bar{a} - \bar{b}\|$. In particular $\bar{a}$ is the projection of $\bar{b}$ onto A, and $\bar{b}$ is the projection of $\bar{a}$ onto B. By projection theorem,

$$\begin{aligned} (\bar{b}-\bar{a})^\top \cdot (a-\bar{a}) &\leq 0 \quad \forall a \in A \\ (\bar{a}-\bar{b})^\top \cdot (b-\bar{b}) &\leq 0 \quad \forall b \in B. \end{aligned}$$

Consider $c = \bar{b} - \bar{a}$. Pick $\bar{x}$ on the segment connection $\bar{a}$ to $\bar{b}$, then

$$H = \bar{x} + \left\{ x : c^\top x = 0 \right\}$$

strictly separates A and B.

■

Proposition 5 (Book 1.5.4)**.** *The closure of the conves hull of a set C is the intersection of all the closed halfspaces containing C.*

Proof.

$$\begin{aligned} H = &\text{ intersection of closed halspaces containing } C \\ = &\bigcap \left\{ \tilde{H} \text{ s.t. } \tilde{H} \text{ is a closed halfspace, } C \subseteq H \right\}. \end{aligned}$$

We prove the double inclusion.

$\mathrm{cl}\,(\mathrm{conv}\,(C)) \subseteq H$**:** if $\tilde{H}$ is a closed subspace containing C, then $\tilde{H}$ contains $\mathrm{cl}\,(\mathrm{conv}\,(C))$, so $\mathrm{cl}\,(\mathrm{conv}\,(C))$ is contained also in their intersection, which is H.

$H \subseteq \mathrm{cl}\,(\mathrm{conv}\,(C))$**:** we prove that $\mathbb{R}^n \setminus \mathrm{cl}\,(\mathrm{conv}\,(C)) \subseteq \mathbb{R}^n \setminus H$. Let $x \in \mathbb{R}^n \setminus \mathrm{cl}\,(\mathrm{conv}\,(C))$. Since $x \notin \mathrm{cl}\,(\mathrm{conv}\,(C))$, by strict separation point 2., we can separate $\{x\}$ from $\mathrm{cl}\,(\mathrm{conv}\,(C))$ via an hyperplane H^*, which is contained in H. The halfspace corresponding to H^* does not contain x, so $x \notin H$. Hence $H \subseteq \mathrm{cl}\,(\mathrm{conv}\,(C))$.

■

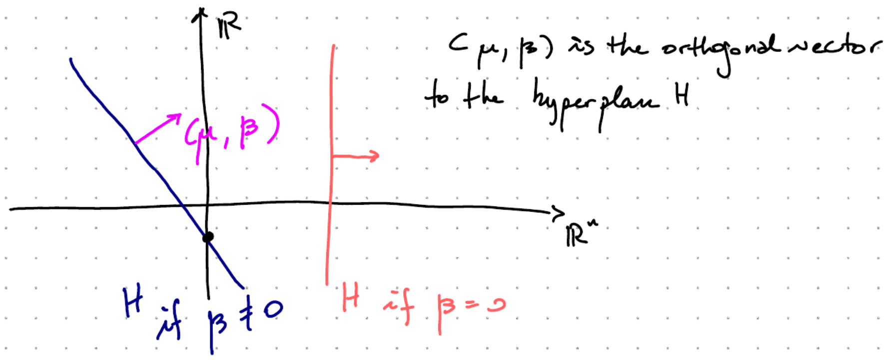
Lecture 12 of 08/04/2026

Let $C \subseteq \mathbb{R}^{n+1}$, $x \notin C$. Idea: if C does not contain vertical lines, we can striclty separate C from x with a "non-vertical" hyperplane.

For $(u, w) \in \mathbb{R}^{n+1}$, $u \in \mathbb{R}^n$, $w \in \mathbb{R}$. An hyperplane H is in the form

$$H = \left\{ (u, w) : (u, w)^\top \cdot (\mu, \beta) = b \right\}$$

for $\mu \in \mathbb{R}^n$, $\beta \in \mathbb{R}$. We say that H is nonvertical if $\beta \neq 0$.



If $(\bar{u}, \bar{w}) \in H$, then $H = (\bar{u}, \bar{w}) + (u, w)$ such that $(u, w) \perp (\mu, \beta)$ (i.e. $(u, w)^\top \cdot (\mu, \beta) = 0$).

Which is the height at which H intersects the $(n+1)$-axis? When the point $(0, \xi) \in H$? If $(0, \xi) = (\bar{u}, \bar{w}) + (u, w)$ for $(u, w) \perp (\mu, \beta)$. Multiply by (μ, β), i.e. take the projection, so

$$\xi\beta = \bar{u}^\top \cdot \mu + \beta\bar{w} + 0 \implies \xi = \frac{\bar{u}^\top \cdot \mu}{\beta} + \bar{w}.$$

Proposition 1 (Non-vertical hyperplane theorem, book 1.5.8). *Let $C \subseteq \mathbb{R}^{n+1}$ nonempty convex set, assume C does not contain any vertical line. Then*

1. *C is contained in the closed halfspace corresponding to a non-vertical hyperplane, i.e there exists $\mu \in \mathbb{R}^n$, $0 \neq \beta \in \mathbb{R}$ and $\gamma \in \mathbb{R}$ such that*

$$\mu^\top u + \beta w \geq \gamma \quad \forall (u, w) \in C;$$

2. *if* $(\bar{u}, \bar{w}) \notin \mathrm{cl}\,(C)$, *there exists a non-vertical hyperplane that striclty separates* $(\bar{u}, \bar{w})$ *and* C.

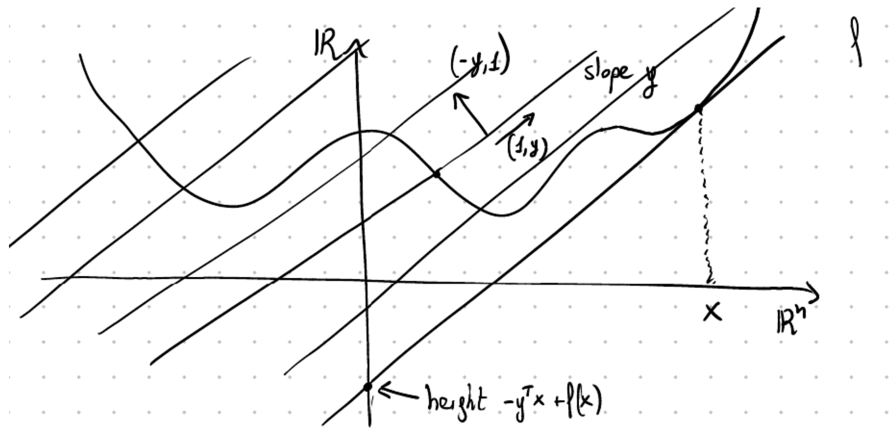
Proof. Not required. ■

12.1 Convex conjugate functions

We would like to describe a funciton f as the supremum of a family of affine functions.

Definition 1. Let $f : \mathbb{R}^n \to \overline{\mathbb{R}}$. We define the *conjugate* of f (or convex conjugate) as the function

$$f^*(y) = \sup_{x \in \mathbb{R}^n} y^\top x - f(x).$$



So what? Assume f given. We want to describe f as sup{affine function below f} = sup{lines of any slope $y \in \mathbb{R}^n$, lines below f}.

Fix a slope $y \in \mathbb{R}^n$ and fix any $(x, f(x))$ in the graph of f. Then the line passing through $(x, f(x))$ and of slope y is

$$\{(x, f(x)) + (u, w) : (u, w) \perp (-y, 1)\}$$

where the "line" $(x, f(x)) + (u, w)$ intersects $(0, \xi)$ at height $-y^\top x + f(x)$.

To keep f above the selected line of slope y, we need to use the one such that the quantity $-y^\top x + f(x)$ is infimized, so $\inf_{x \in \mathbb{R}^n} -y^\top x + f(x)$ is the intercept of the "highest" line with slope y that passes from a point in the graph of f and contains the graph of f in one of its closed halfspaces. Now we observe that

$$\inf_{x \in \mathbb{R}^n} -y^\top x + f(x) = \sup_{x \in \mathbb{R}^n} y^\top x - f(x) = -f^*(y).$$

Definition 2 (Biconjugate function)**.** The biconjugate funciton is defined as

$$f^{**}(x) = \sup_{y \in \mathbb{R}^n} x^\top y - f^*(y).$$

Example 3. Let $f(x) = x^2$. We want to compute its conjugate f^*.

From the definition, $f^*(y) = \sup_{x \in \mathbb{R}^n} \{yx - f(x)\}$. One strategy can be find the critical points of the function $g(x) = yx - x^2 \implies g'(x) = y - 2x \implies x = y/2$ is a critical point (in particular a maximum). Hence

$$f^*(y) = y\frac{y}{2} - (\frac{y}{2})^2 = \frac{y^2}{4}.$$

Example 4. Let $f(x) = e^x$. Again, we can compute $g'(x)$ and find a critical point:

$$g(x) = yx - e^x \implies g'(x) = y - e^x \implies y = e^x \text{ critical point.}$$

Now, if $y > 0$ we are fine, since $x = \log(y)$. Otherwise? If $y < 0$ we would have $yx - e^x =$ negative $\cdot x - e^x$ which explodes to $+\infty$ as $x \to -\infty$. If instead $y = 0$, $\sup_x -e^x = 0$. Finally,

$$f^*(y) = \begin{cases} y\log(y) - y & y > 0 \\ 0 \quad y = 0 \\ +\infty \quad y < 0 \end{cases}.$$

Example 5. Consider $f(x) = x - 1$. Then

$$f^*(y) = \sup_{x \in \mathbb{R}} yx - x + 1 = x(y-1) + 1.$$

If $y \neq 1$, then we can get $x \to +\infty$ and we explode, hence

$$f^*(y) = \begin{cases} 1 & \text{if } y = 1 \\ +\infty & \text{otherwise.} \end{cases}$$

Example 6. Consider $f(x) = |x|$. Define

$$g(x) = \begin{cases} yx + x & \text{if } x \leq 0 \\ yx - x & \text{if } x \geq 0 \end{cases} = \begin{cases} x(y+1) & \text{if } x \leq 0 \\ x(y-1) & \text{if } x \geq 0 \end{cases}$$

We have 3 cases:

$$\begin{aligned} y + 1 < 0 &\implies x(y+1) \nearrow +\infty \text{ as } x \searrow -\infty \\ y - 1 > 0 &\implies x(y-1) \nearrow +\infty \text{ as } x \nearrow -\infty. \end{aligned}$$

If instead $|y| \leq 1$ we have

$$g(x) = \begin{cases} x \cdot \text{ positive} & x \leq 0 \\ x \cdot \text{ negative} & x \geq 0 \end{cases}$$

In both cases, the supremum is 0. Finally,

$$f^*(y) = \begin{cases} 0 & \text{if } |y| \leq 1 \\ +\infty & \text{otherwise} \end{cases} = \delta_{\{y:|y|\leq 1\}}$$

the convex indicator function of the set $\{y : |y| \leq 1\}$.

Chapter 13

Lecture 13 of 14/04/2026

Proposition 1 (Conjugate function theorem, Book 1.6.1). *Let $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ and denote with f^* its conjugate, $f^{**} = (f^*)^*$. Then:*

a) *$f(x) \geq f^{**}(x)$ for any $x \in \mathbb{R}^n$;*

b) *if f is convex, then if one between f, f^*, f^{**} is proper, then also the other two are;*

c) *if f is closed and convex, $f(x) = f^{**}(x)$ for every $x \in \mathbb{R}^n$;*

d) *$f^* = \left(\check{\mathrm{cl}}\,(f)\right)^*$. Furthermore, if $\check{\mathrm{cl}}\,(f)$ is proper, then $\check{\mathrm{cl}}\,(f) = f^{**}$.*

Proof. <u>**a)**</u>: $f^*(y) = \sup_x y^\top x - f(x)$. This means $f^*(y) \geq y^\top x - f(x)$ for any y, x, this implies $y^\top x \leq f(x) + f^*(y)$ (this last is called Fenchel's inequality). Now,

$$f^{**}(x) = \sup_y x^\top y - f^*(y) \leq f(x).$$

<u>**b)**</u>: assume f is convex and proper, so $\mathrm{epi}\,(f)$ is nonempty convex set in $\mathbb{R}^{n+1}$, since f is proper, $\mathrm{epi}\,(f)$ does not contain any vertical line, so there exists $(y, 1) \in \mathbb{R}^{n+1}$, $c \in \mathbb{R}$ such that $y^\top x + f(x) \geq c$ for any $x \in \mathbb{R}^n$ (because the epigraph lies above the tangent hyperplane). This implies $f^*(-y) \leq -c \implies \sup_x -y^\top x - f(x) \leq -c$, so $f^*(y) < +\infty$. Is $f^* \cong -\infty$? No, indeed there exists $\bar{x}$ such that $f(\bar{x})$ is finite (f is proper), hence $f^*(y) = \sup_x y^\top x - f(x) \geq t^\top x - f(\bar{x})$. This shows that f is proper (the same computation holds for the others).

<u>**c)**</u>: from a) we know that $f^{**}(x) \leq f(x)$. We wanto to prove $f^{**}(x) \geq f(x)$, so $\mathrm{epi}\,(f^{**}) \subseteq \mathrm{epi}\,(f)$. By contraddiction, assume $x \in \mathrm{epi}\,(f^{**}) \setminus \mathrm{epi}\,(f)$. Then we can separate a point $\{(x, \gamma)\}$ form $\mathrm{epi}\,(f)$, also strictly, so there exists (y, ξ), $\xi \neq 0$, $c \in \mathbb{R}$ such that

$$y^\top z + \xi w < c < y^\top x + \xi \gamma$$

for any $(z, w) \in \mathrm{epi}\,(f)$. Observe that we can take $w \nearrow +\infty$, so we must have $\xi < 0$. Assume WLOG $\xi = -1$ (by scaling). So

$$y^\top z - f(z) < c < y^\top x - f^{**}(x) \quad \forall x \in \mathrm{dom}\,(f).$$

Take the supremum over z, so

$$\sup_z f^*(y) = \sup_z y^\top z - f(x) < y^\top - f^{**}(x)$$
$$\implies y^\top x > f^*(y) + f^{**}(x)$$

which contraddicts the Fenchel's inequality.

d): define $\check{f}^* = \left(\check{\mathrm{cl}}\,(f)\right)^*$. Then $\check{\mathrm{cl}}\,(f)$ is a convex closed function, so epi $\left(\check{\mathrm{cl}}\,(f)\right) =$ cl (conv (epi (f))). We know cl (conv (epi (f))) = intersection of all the closed halfspaces that contains epi (f). This implies that $-f^*(y)$ is intercept of the highest hyperplane such that epi (f) is above that hyperplane, hence $-f^*(y) = \check{f}^*(y)$ because hyperplanes below f and $\check{\mathrm{cl}}\,(f)$ are the same. This shows $f^* = \check{f}^*$, then $f^{**} = \left(\check{\mathrm{cl}}\,(f)\right)^* = \check{\mathrm{cl}}\,(f)$ by point c).

■

13.1 Convex optimization

We move now the focus onto the actual optimization of convex functions. We set ourselfs on the problem

$$\inf_{x\in X} f(x)$$

where $X \subseteq \mathbb{R}^n$, $f : X \to (-\infty, +\infty]$. We say that x^* is a minimum point if it is such that

$$x^* = \mathrm{argmin}_{x\in X} f(x).$$

If $x \in X \cap \mathrm{dom}\,(f)$, then $f(x^*) = \inf_{x\in X} f(x) = \min_{x\in X} f(x)$, so the infimum is attained as x^*.

We recall tha we say that x^* is a

- local minima if $f(x^*) \leq f(x)$ for any $x \in B(x^*, \varepsilon)$ for some $\varepsilon > 0$;
- global minima if $f(x^*) \leq f(x)$ for any $x \in X$.

Proposition 1 (Book 3.1.1). *Let $X \subseteq \mathbb{R}^n$ be convex, $f : X \to (-\infty, +\infty]$ be a convex function. Then local minima are global minima. In addition, if f is strictly convex, then there exists at most one global minima.*

Proof. Assume x^* local minima but not global, i.e. there exists $\bar{x}$ such that $f(\bar{x}) < f(x^*)$. Then

$$f(\alpha x^* + (1-\alpha)\bar{x}) \le \alpha f(x^*) + (1-\alpha) f(\bar{x}) < f(x^*) \quad \forall \alpha \in [0,1].$$

For $\alpha \to 1$, $\alpha x^* + (1-\alpha)\bar{x} \in B(x^*, \varepsilon)$, where ε is such that x^* is local minima in $B(x^*, \varepsilon)$, which contraddicts the fact that x^* is local minima, hence it is also a global one.

Furthermore, if f is striclty convex, assume x, x^* two global minima, so $f(x) = f(x^*)$. By strict convexity,

$$f\left(\frac{1}{2}x + \frac{1}{2}x^*\right) < \frac{1}{2}f(x) + \frac{1}{2}f(x^*) = f(x).$$

This shows that $x = x^*$.

■

The natural question now is: does a minimum point exists? We will see that under some condition this existance is guaranteed.

Definition 2 (Coercivity)**.** A function $f : \mathbb{R}^n \to (-\infty, +\infty]$ is *coercive* if for every sequence $\{x_k\}_k \subseteq \mathbb{R}^n$ such that $\|x_k\| \to +\infty$, we have

$$\lim_{k \to +\infty} f(x_k) \to +\infty.$$

Proposition 3 (Weierstrass, Book 3.2.1)**.** *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ closed proper and LSC. Assume one among the following:*

1. $\mathrm{dom}\,(f)$ *is bounded;*
2. *there exists $\bar{\gamma}$ such that the set $\{x : f(x) \le \bar{\gamma}\}$ is nonempty and bounded;*
3. *f is coercive.*

Then there exists $x^* \in \mathbb{R}^n$ such that $f(x^*) \le f(x)$ for any $x \in \mathbb{R}^n$, i.e. x^* is a global minimum point.

Proof. Let $\{x_k\}_k$ be a minimizing sequence for f, so $f(x_k) \to \inf_x f(x)$. Then $\{x_k\}_k \subseteq \mathrm{dom}\,(f)$.

1. If 1., $\{x_k\}_k \subseteq$ bounded set;
2. if 2., $f(x_k) \le \bar{\gamma}$ for k big enough, hence $\{x_k\}_k$ is bounded;
3. if 3., $\{x_k\}_k$ cannot "go to ∞", otherwise $f(x_k) \to +\infty$ since f is coercive.

In any of these, the sequence $\{x_k\}_k$ is *bounded*. By BW, up to subsequences, $x_k \to \bar{x} \in \mathbb{R}^n$. Using the closerness and lower semicontinuity of f, it holds true that

$$f(\bar{x}) \leq \liminf_{k \to +\infty} f(x_k) = \inf_x f(x)$$

hence $\bar{x}$ is a global minimum point.

■

Proposition 4 (Book 3.2.2)**.** *Let $X \subseteq \mathbb{R}^n$ closed convex, $f : \mathbb{R}^n \to (-\infty, +\infty]$ convex closed, assume $X \cap \operatorname{dom}(f) \neq \emptyset$. Then the set of minima of f over X is nonempty if and only if X and f has no nonzero direction of recession.*

Proof. Assume $\tilde{f} = \inf_{x \in X} f(x)$. Notice that $\tilde{f} < +\infty$, because $X \cap \operatorname{dom}(f)$ is nonempty. Consider $\{\gamma_k\}_k$ such that $\gamma_k \to \tilde{f}$, $\gamma_k \geq \tilde{f}$ for any k. Define $V_k = \{x : f(x) \leq \gamma_k\}$. Then the minimum point belongs to the intersection $\bigcap_k (X \cap V_k)$. The problem now is to guarantee that the latter intersection is nonempty. In fact it is nonempty, indeed

$$\mathrm{R}_X \cap \mathrm{R}_{V_k} = \mathrm{R}_X \cap \mathrm{R}_f = \{0\} .$$

By theorem 1.4.2(c) of the book, this implies the intersection nonempty.

■

CHAPTER 14

LECTURE 14 OF 15/04/2026

Example 1. Consider $f(x) = e^{-x}$, $X = (-\infty, 1]$. Then f is convex and closed. Then

$$\begin{aligned} \mathrm{r}_f &= \{d : d \geq 0\} \\ \mathrm{R}_X &= \{d : d \leq 0\}. \end{aligned}$$

Hence $\mathrm{r}_f \cap \mathrm{R}_X = \{0\}$, this implies that f attains the infimum over X.

PROPOSITION 2 (Book 3.2.4). *Let X be closed convex in $\mathbb{R}^n$, $f : \mathbb{R}^n \to (-\infty, +\infty]$ convex and closed. Assume $X \cap \mathrm{dom}\,(f)$ nonempty. Then the set of minima of f over X, denoted by X^* is nonempty under one among the following assumptions/conditions:*

1. $\mathrm{R}_f \cap \mathrm{R}_X = \mathrm{L}_X \cap \mathrm{L}_f$;
2. $\mathrm{R}_X \cap \mathrm{R}_f \subseteq \mathrm{L}_f$ *and X is polyhedral.*

Furthermore, under 1., $X^* = \tilde{X} + (\mathrm{L}_x \cap \mathrm{L}_f)$ where $\tilde{X}$ is some compact set in $\mathbb{R}^n$.

Proof. Let $\tilde{f} = \inf_{x \in X} f(x) < +\infty$ as $X \cap \mathrm{dom}\,(f)$ is nonempty. Let $\{\gamma_k\}_k$ such that $\gamma_k \to \tilde{f}$ and $\gamma_k \geq \tilde{f}$ for any k. Set $V_k = \{x : f(x) \leq \gamma_k\}$. Note $X^* = \bigcap_k (X \cap V_k)$.

Assume 1., then $\{X \cap V_k\}_k$ is a nestes sequence of closed convex sets. $\mathrm{R}_X \cap V_k = \mathrm{R}_X \cap \mathrm{R}_{V_k} = \mathrm{R}_X \cap \mathrm{R}_f = \mathrm{L}_X \cap \mathrm{L}_f = \mathrm{L}_X \cap \mathrm{L}_{V_k} = \mathrm{L}_{X \cap V_k}$. By 1.4.11(a) X^* is nonempty and it holds ist characterization.

Assume 2., then $\mathrm{R}_X \cap \mathrm{R}_{V_k} = \mathrm{R}_X \cap \mathrm{R}_f \subseteq \mathrm{L}_f$, X polyhedral, so retractive. By 1.4.11(b), X^* is nonempty. ∎

Chapter 15

Lecture 15 of 21/04/2026

15.1 Minmax problems

Let $X, Y \subseteq \mathbb{R}^n$, $L : X \times Y \to \overline{\mathbb{R}}$. Define

$$f(x) = \sup_y L(x,y)$$
$$g(y) = \inf_x L(x,y).$$

So $f : X \to \mathbb{R}$, $g : Y \to \mathbb{R}$. We are intrested in $\inf_x f(x)$ and $\sup_y g(y)$. How do they relate?

Remark 1. By definition, $g(y) \leq L(x,y) \leq f(x)$ for anu $x \in X$ and $y \in Y$. In particular, $g(y) \leq f(x)$. Thake the $\inf_x \sup_y$ on both sides, so

$$\sup_y g(y) \leq \inf_x f(x).$$

We get then $\sup_y \inf_x L(x,y) \leq \inf_x \sup_y L(x,y)$.

Definition 2 (Saddle value)**.** If $\sup_y \inf_x L(x,y) = \inf_x \sup_y L(x,y)$ we name this value *saddle value* of L.

Definition 3 (Saddle point)**.** A point $(\bar{x}, \bar{y})$ is a *saddle point* of L if

$$L(\bar{x}, y) \leq L(\bar{x}, \bar{y}) \leq L(x, \bar{y})$$

for any x, y.

Theorem 4. *A pair $(\bar{x}, \bar{y})$ is a saddle point if and only if $\bar{x}$ solves $\inf_x f(x)$, $\bar{y}$ solves $\sup_y g(y)$ and the saddle value exists, i.e. $\inf_x f(x) = \sup_y g(y)$.*

Proof. $\underline{'' \implies ''}$: suppose $(\bar{x}, \bar{y})$ saddle point. Then $L(\bar{x}, y) \leq L(\bar{x}, \bar{y}) \leq L(x, \bar{y})$ for any x, y. This implies

$$\sup_y L(\bar{x}, y) \leq L(\bar{x}, \bar{y}) \leq \inf_x L(x, \bar{y}),$$

hence,

$$f(\bar{x}) \leq L(\bar{x}, \bar{y}) \leq g(\bar{y}) \leq f(\bar{x})$$

as $g \leq f$, so the inequalites becomes actually equalities.
The other implication is easy.

■

Why are we intrested in this kind of problems? Our focus is $\inf_x f(x)$. Assume, by chance, that such an L exists and be such that $f(x) = \sup_y L(x, y)$. Now, how can we find Y and L?

In this setting i can look at $\sup_y g(y)$ as a natural candidate of the "dual" problem, and hope that solving the dual is easier. We know for sure $g(y) \leq \inf_x f(x)$ for any y, hence $g(y)$ is a lower bound on the infimum in the primal problem.

Assume that exists $\bar{y}$ such that $g(\bar{y}) = \inf_x f(x)$. But also $L(x, \bar{y}) \leq f(x)$ for any x. So if $\bar{x}$ minimizes f, then $\bar{x}$ also minimizes $L(\cdot, \bar{y})$. So we can determine all minimizers of $L(\cdot, \bar{y})$ and search among them a minimizer of f. Observe that $g(\bar{y}) \leq \sup_y g(y) \leq \inf_x f(x) = g(\bar{y})$. Then $\bar{y}$ maximizes $g(y)$, hence $(\bar{x}, \bar{y})$ is a saddle point of L. But finding L is not easy…

But under conveity assumptions, all this process could be implemented easly. Let's focus on the convex case. The idea is to "embed" the $\inf_x f(x)$ into a "bigger" function F, so define $F : X \times U \to \overline{\mathbb{R}}$, where U is a set of perturbations, select F such that $F(x, 0) = f(x)$.

Definition 5 (Optimal value function)**.** The *optimal value function* $\varphi : U \to \overline{\mathbb{R}}$ is defined as

$$\varphi(u) = \inf_x F(x, u).$$

Observe that $\varphi(0) = \inf_x f(x)$.

Theorem 6 (Book 3.3.1(a))**.** *If F is convex (in both entries ath the same time), then φ is convex.*

Proof. If $\operatorname{epi}(\varphi) = \emptyset$, then $\varphi \equiv +\infty$, which is convex. Otherwise, is $\operatorname{epi}(\varphi)$ convex? Let $(\bar{u}, \bar{v}), (\tilde{u}, \tilde{v}) \in \operatorname{epi}(\varphi)$. Then

$$\inf_x F(x, \bar{u}) = \varphi(\bar{u}) \leq \bar{v}$$

so there exists a sequence $\{\bar{x}_k\}_k \subseteq X$ such that $F(\bar{x}_k, \bar{u}) \to \varphi(\bar{u})$. We do the same reasoning for , so there exists a sequence $\{\tilde{x}_k\}_k$ such that $F(\tilde{x}_k, \tilde{u}) \to \varphi(\tilde{u})$. Take $\alpha \in (0, 1)$, then

$$\varphi(\alpha\bar{u} + (1-\alpha)\tilde{u}) \leq F(\alpha\bar{x}_k + (1-\alpha)\tilde{x}_k, \alpha\bar{u} + (1-\alpha)\tilde{u}).$$

Using the convexity of F (in both arguments), we get

$$\begin{array}{cc} \alpha F(\bar{x}_k, \bar{u}) & +(1-\alpha)F(\tilde{x}_k, \tilde{u}) \\ \downarrow & \downarrow \\ \varphi(\bar{u}) & \varphi(\tilde{u}). \end{array}$$

Going to the limit,

$$\begin{aligned} \varphi(\alpha\bar{u} + (1-\alpha)\tilde{u}) &\leq \alpha\varphi(\bar{u}) + (1-\alpha)\varphi(\tilde{u}) \\ &\leq \alpha\bar{v} + (1-\alpha)\tilde{v}. \end{aligned}$$

Therefore, $(\alpha\bar{u} + (1-\alpha)\tilde{u},\ \alpha\bar{v} + (1-\alpha)\tilde{v}) \in \text{epi}\,(\varphi)$.

■

Example 7. Fix $f_0, f_1, \ldots, f_m$ convex functions on C nonempty convex. Define

$$f(x) = \begin{cases} f_0(x) & \text{if } x \in C,\ f_i(x) \leq 0\,\forall i = 1, \ldots, m \\ +\infty & \text{otherwise} \end{cases}$$

Where we put the perturbations? The idea is to perturb the constraints, therefore

$$F(x,u) = \begin{cases} f_0(x) & \text{if } x \in C,\ f_i(x) \leq u_i\,\forall i = 1, \ldots, m \\ \\ +\infty & \text{otherwise} \end{cases}$$

Then $F : C \times U \to \overline{\mathbb{R}}$, and $f(x) = F(x, 0)$. Moreover, F is convex both in x and u.

Example 8. Same structure as above. Assume furthermore that

$$f_i(x) = h_i(A_i x + a_i) + \ell_i(x)$$

where $A_0, \ldots, A_m$ are matrices, $h_i : Z_i \to \overline{\mathbb{R}}$, $a_i \in Z_i$ and ℓ_i are affine functions for every i.

Then $u = (u_1, \ldots, u_m, z_1, \ldots, z_m) \in \mathbb{R}^m \times Z_0 \times \cdots \times Z_m =: U$. Then

$$F(x,u) = \begin{cases} h_0(A_0 x + a_0 - z_0) + \ell_0(x) & \text{if } x \in C,\ h_i(A_i x + a_i - z_i) + \ell_i(x) \leq u_i \\ \\ +\infty & \text{otherwise} \end{cases}$$

Example 9. Consider the first example setting. Another possibility is to choose

$$F(x,u) = \begin{cases} f_0(x) + r\|u\|^2 & \text{if } x \in C, \ , f_i(x) \le u_i \\ \\ +\infty & \text{otherwise} \end{cases}$$

Definition 10 (Concave conjugate)**.** Let g be concave. We define the concave conjugate as

$$g_*(y) = \inf_x y^\top x - g(x).$$

We define the *Lagrangian* L associated to the representation F the function

$$L(x,y) = \inf_u F(x,u) + y^\top u = (-F(x,\cdot))_* \,(y).$$

Assume $F(x,\cdot)$ closed convex for every fixed x. Then

$$L(x,y) = \Bigg(\underbrace{F(x,\cdot)}_{\text{concave closed}} \Bigg)_* (y) \implies (L(x,\cdot))_* = -F(x,\cdot).$$

Thus

$$\begin{aligned} -F(x,u) &= \inf_y y^\top u - L(x,y) \\ F(x,u) &= \sup_y L(x,y) - y^\top u \end{aligned}$$

if we assume $F(x,\cdot)$ convex closed.

Example 11. Set as the first example and consider $L(x,y) = \inf_u F(x,u) + y^\top u$. If $x \notin C$ we get $+\infty$. If some $y_i < 0$, then we can take u as large as we want and we get $-\infty$. Assume then $x \in C$, $y_i \ge 0$ for every i. Then

$$L(x,y) = \inf_{u: f_i(x) \le u_i} f_0(x) + y^\top u.$$

We have to take the "smaller" possible u, thus $u_i = f_i(x)$. Under this condition, we get

$$L(x,y) = f_0(x) + \sum_{i=1}^m y_i f_i(x)$$

which is the standard Lagrangian in $\mathbb{R}^n$, therefore we are very happy.

CHAPTER 16

LECTURE 16 OF 22/04/2026

THEOREM 1. *Book ???*

1. *$L(x,\cdot)$ is closed and concave (the hypergraph below is closed and it is upper semi-continous, i.e. $\{(y,w) : w \leq L(x,y)\}$);*
2. *if $F(x,\cdot)$ is closed convex for every fixed x, then $f(x) = \sup_y L(x,y)(\star)$;*
3. *if $L : X \times Y \to \overline{\mathbb{R}}$ is a function such that $(\star)$ holds, and if $L(x,\cdot)$ is closed concave for every x, then L is the Lagrangian associated with a uniquely determined representation $f(x) = F(x,0)$ having F closed convex in u, with $F(x,u) = \sup_y L(x,y) - y^\top u$;*
4. *if $F(x,\cdot)$ is closed convex, then $L(\cdot,y)$ is convex for every fixed y if and only if F is convex in (x,u).*

Proof. **1.**: $L(x,y) = \inf_u F(x,u) + y^\top u = \inf_u$ affine function in y, which is closed and concave.

2.: we proved it last lecture, so

$$F(x,u) = \sup_y L(x,y) - y^\top u$$

but $f(x) = F(x,0) = \sup_y L(x,y)$.

3.: follows from the definition.

4.: assume $F(x,\cdot)$ closed convex. Assume $L(\cdot,y)$ convex. Then the map $(x,u) \mapsto L(x,y) - y^\top u$ is convex in x for each fixed y. It is also convex in u for any y fixed. From 2., $F(x,u) = \sup_u L(x,y) - y^\top u$, which is convex as the supremum of a family of jointly convex functions in x and u.

Assume now F jointly convex in (x,u). Then

$$L(x,y) = \inf_u F(x,u) + y^\top u$$

which, for any fixed y, is a convex function in (x,u). Therefore, $L(\cdot,y)$ is convex as optimal value function.

■

From now on, we will denote with

$$
\begin{aligned}
&(P)\ \inf_x f(x) \text{ where } f(x) = \sup_y L(x,y) \text{ as the primal problem}\\
&(D)\ \sup_y g(y) \text{ where } g(y) = \inf_x L(x,y) \text{ as the dual problem}
\end{aligned}
$$

and the optimal value function $\varphi(x) = \inf_u F(x,u)$.

Theorem 2 (Main). 1. *The function g in (D) is concave and closed. Indeed we have the following relation:*

$$g = (-\varphi)_*,\ -g_* = \operatorname{cl}(\operatorname{co}(\varphi))$$

the convex closure of φ;

2. *if F is jointly convex in (x,u) then $-g_* = \operatorname{cl}(\varphi)$ and, indeed,*

$$\sup_y g(y) = \liminf_{u\to 0} \varphi(u)$$

assuming $0 \in \operatorname{cl}(\operatorname{dom}(\varphi))$.

Proof.

$$
\begin{aligned}
g(y) &= \inf_x L(x,y)\\
&= \inf_{x,u} F(x,u) + y^\top u\\
&= \inf_u \inf_x F(x,u) + y^\top u\\
&= \inf_u \varphi(u) + y^\top u\\
&= \inf_u y^\top u - (-\varphi)(u) = (-\varphi)_*(y).
\end{aligned}
$$

Furthermore,

$$g = (-\varphi)_* \implies g_* = (-\varphi)_{**}.$$

Observing that $-(-\varphi)_{**} = \varphi^{**}$ we conclude that $-g_* = \varphi^{**} = \check{\operatorname{cl}}(\varphi)$.

If F is jointly ocnvex in (x,u) we know that φ is convex, hence $\check{\operatorname{cl}}(\varphi) = \operatorname{cl}(\varphi)$. Since we know that the closure of φ is LSC, then $\sup_y g(y) = \inf_x f(x) = \operatorname{cl}(\varphi)(0) = \liminf_{u\to 0}\varphi(u)$.

■

How if we reverse the problem? Consider $g(y) = G(y, 0)$, so that $G : Y \times V \to \overline{\mathbb{R}}$, $G(y, v) = \inf_x L(x, y) - v^\top x$. Define the *dual optimal value function* $\gamma(v) = \sup_y G(y, v)$. Observe that

$$G(y, v) = \inf_x \inf_u F(x, u) + y^\top u - v^\top x = -F^*(v, -y)$$

where

$$F^*(v, -y) = \sup_{x,u} (v, -y)^\top (x, u) - F(x, u).$$

Theorem 3. *1. G concave closed, γ closed;*

2. if F is convex (jointly in (x, u)) then

$$F(x, u) = \sup_{y,v} G(y, v) + x^\top v - y^\top u = -G^*(-x, u).$$

Moreover, $f = (-\gamma)^*$, $-f^* = \mathrm{cl}\,(\gamma)$;

3. if f is convex, then

$$\inf_x f(x) = \limsup_{v \to 0} \gamma(v)$$

assuming $0 \in \mathrm{cl}\,(\mathrm{dom}\,(\gamma))$.

Exercise.

$$\begin{aligned} F(x, u) &= \sup_y L(x, y) - y^\top u \\ &= \sup_{y,v} G(y, v) + x^\top v - y^\top u \\ &= \sup_{y,v} -(-G(y, v)) - y^\top u - (-x)^\top v = -G^*(-x, u). \end{aligned}$$

Same argument as before for the rest.

■

Example 4. Consider the setting $f_0, f_1, \ldots, f_m$, the objective is

$$\inf_{x \in C} f_0(x) \text{ such that } f_i(x) \le 0.$$

We already computed $F(x, u)$ and $L(x, y)$. Then

$$g(y) = \inf_x L(x, y) = \begin{cases} \inf_x \; f_0(x) + \sum_i y_i f_i(x) & \text{if } y_i \ge 0 \\ \\ -\infty \quad \text{otherwise} \end{cases}$$

Let's consider a subcase: the linear programming problem. We have

$$f_0(x) = \langle b_0, x\rangle + \beta_0,\ b_0 \in \mathbb{R}^n,\ \beta_0 \in \mathbb{R}$$

$$f_i(x) = \langle b_i, x\rangle + \beta_i,\ b_i \in \mathbb{R}^n,\ \beta_i \in \mathbb{R}\,\forall i.$$

it is usually presented as

$$\inf c^\top x$$

$$Ax \leq b.$$

We seek for $g(y)$ such that $g(y) = G(y, 0)$. Who is $G(y, v)$? Recall that $L = f_0 + \sum_i y_i f_i$. Then

$$\begin{aligned} G(y,v) &= \inf L(x,y) - v^\top x \\ &= \inf_x \langle b_0, x\rangle + \beta_0 + \sum_{i=1}^m y_i \left[\langle b_i, x\rangle + \beta_i\right] - v^\top x \\ &= \inf_x \beta_0 + \sum_i y_i\beta_i + \langle x, b_0 + \sum_i y_i b_i - v\rangle \\ &= \begin{cases} \beta_0 + \sum_{i=1}^m y_i\beta_i & \text{if } v = b_0 + \sum_i y_i b_i \\ \\ -\infty \quad \text{otherwise} \end{cases} \end{aligned}$$

We imposed that $b_0 + \sum_i y_i b_i - v = 0$ because if we took $x \searrow -\infty$, then $\langle x, b_0 + \sum_i y_i b_i - v\rangle \searrow -\infty$, therefore the infimum would be $-\infty$. Therefore the dual problem turns out to be

$$\sup_{y \geq 0} \beta_0 + \sum_{i=1}^m y_i\beta_i$$
$$\beta_0 + \sum_{i=1}^m y_i b_i = 0.$$

Remark 5. Observe that F jointly convex in (x, u) implies φ convex. Then also epi (φ) = projection of epi (f) through the map $(x, u, \alpha) \mapsto (u, \alpha)$. Observe that epi (F) is polyhedral, indeed is the intersection of closed halfspaces, as each f_i is an affine function. Then epi (φ) is a linear map applied to a polihedral set. This shows that epi (φ) is closed (as seen in the previous lecture).

Chapter 17

Lecture 17 of 28/04/2026

17.1 Fenchel's duality

Consider

$$
\begin{aligned}
h :& \mathbb{R}^n \to \overline{\mathbb{R}} \text{ proper convex}\\
K :& \mathbb{R}^m \to \overline{\mathbb{R}} \text{ proper convex}\\
A :& \mathbb{R}^n \to \mathbb{R}^m \text{ linear application}
\end{aligned}
$$

and set the problem to be

$$\inf_x f(x),\ f(x) = h(x) + K(Ax).$$

Then the preturbation function is

$$F(x,u) = h(x) + K(Ax+u).$$

The Lagrangian is

$$
\begin{aligned}
L(x,y) &= \inf_u F(x,u) + y^\top u\\
&= \inf_u h(x) + K(Ax+u) + y^\top\\
&= h(x) + \inf_u K(Ax+u) + y^\top u \quad \tilde{u} = Ax+u\\
&= h(x) + \inf_{\tilde{u}} K(\tilde{u}) + y^\top(\tilde{u} - Ax)\\
&= h(x) - \sup_{\tilde{u}} (-y)^\top \tilde{u} - K(\tilde{u}) - y^\top Ax\\
&= h(x) - K^*(-y) - y^\top Ax
\end{aligned}
$$

assuming $h(x) < +\infty$.
In the dual, consider the perturbation function

$$\begin{aligned} G(y,v) &= -F^*(v,-y) \\ &= -\sup_{(x,u)} v^\top x - y^\top u - h(x) - K(Ax+w) \\ &= -\sup_{(x,\tilde{u})} v^\top x - y^\top \tilde{u} + y^\top Ax - h(x) - K(\tilde{u}) \\ &= -\sup_{x} \underbrace{v^\top x + y^\top Ax}_{=(v+A^\top y)^\top x} - h(x) - \sup_{\tilde{u}} (-y)^\top \tilde{u} - K(\tilde{u}) \\ &= -h^*(v + A^\top y) - K^*(-y). \end{aligned}$$

So $g(y) = G(y,0) = -h^*(A^\top y) - K^*(-y)$, hence the dual problem is

$$\sup_y -h^*(A^\top y) - K^*(-y).$$

17.2 Subdifferentials

Recall 1. If $\varphi : U \to \overline{\mathbb{R}}$ is convex, then φ is continous on $\operatorname{int}(\operatorname{dom}(f))$

Recall 2. $0 \in \operatorname{int}(\operatorname{dom}(f)) \iff \{y : \varphi^*(y) \leq \beta\}$ is compact for any β and one subset of this type is nonempty, $\varphi(0) < +\infty$.

Recall 3. Let φ convex. If one of the sublevel sets is compact, then all sublevel sets are compact.

Recall 4. For any φ, the convex conjugate function $\varphi^*(y) = \sup_u y^\top u - \varphi(u)$ is convex.

What about differentiability? In general, convexity does not imply differentiability, e.g. $\varphi(x) = |x|$.

Definition 1 (Subgradient/subdifferential)**.** Let $\varphi : U \to \overline{\mathbb{R}}$, $u \in U$. We say that $y \in U$ is a *subgradient* of φ at u if

$$\varphi(v) \geq \varphi(u) + \langle y, v-u \rangle$$

for any $v \in U$. Furthermore denote as the *subdifferential* of φ at u the set

$$\partial\varphi(u) = \{y : y \text{ subgradient of } \varphi \text{ at } u\}.$$

Remark 2. If φ is differentiable at u, then $\partial\varphi(u) = \{\nabla\varphi(u)\}$.

Assume now $\varphi : U \to \overline{\mathbb{R}}$ convex proper, $\varphi \neq +\infty$.
closed and convex: consider $y_1, y_2 \in \partial\varphi(u)$, $\alpha \in (0,1)$. Then

- $\partial\varphi(u)$ closed and convex:

$$\varphi(v) \geq \varphi(u) + \langle y_1, v-u\rangle \;\forall v$$
$$\varphi(v) \geq \varphi(u) + \langle y_2, v-u\rangle \;\forall v.$$

  Multiply the first by α and second by $(1-\alpha)$. Then

$$\alpha\varphi(v) + (1-\alpha)\varphi(v) \geq \alpha\varphi(u) + (1-\alpha)\varphi(u) + \langle \alpha y_1 + (1-\alpha)y_2, v-u\rangle$$
$$\varphi(v) \geq \varphi(u) + \langle \alpha y_1 + (1-\alpha)y_2, v-u\rangle \;\forall v.$$

  Therefore, $\alpha y_1 + (1-\alpha)y_2 \in \partial\varphi(u)$. Consider now a sequence $\{y_k\}_k \subseteq \partial\varphi(u)$ such that $y_k \to \bar{y}$. Then

$$\begin{array}{lll} \varphi(v) & \geq \varphi(u) & +\langle y_k, v-u\rangle \\ & & \quad\downarrow \\ \varphi(v) & \geq \varphi(u) & +\langle \bar{y}, v-u\rangle \end{array}$$

  Hence $\bar{y} \in \partial\varphi(u)$.

- If φ is continous at u, then $\partial\varphi(u)$ is bounded: let $y \in \partial\varphi(u)$, then $\varphi(v) \geq \varphi(u) + \langle y, v-u\rangle$. By continuity of φ in u, there exists $\delta > 0$ such that $|\varphi(v) - \varphi(u)| \leq 1$ for any $v \in B(u,\delta)$. Then

$$\langle y, v-u\rangle \leq |\varphi(v) - \varphi(u)| \leq 1 \implies \sup_{\|z\|\leq\delta} \langle y, z\rangle \leq 1.$$

  We know that $\|y\| = \sup_{\|x\|\leq 1}\langle x, y\rangle$, therefore

$$\delta\|y\| \leq 1 \implies \|y\| \leq 1/\delta$$

  showing that $\partial\varphi(u)$ is bounded.

- $\varphi(\bar{u}) = \inf_u \varphi(u) \iff 0 \in \partial\varphi(\bar{u})$:

  $0 \in \partial\varphi(\bar{u}) \implies \varphi(v) \geq \varphi(\bar{u}) + \langle 0, v-\bar{u}\rangle \implies \varphi(v) \geq \varphi(\bar{u}) \implies \bar{u}$ is a min
  $\varphi(\bar{u}) = \inf\limits_u \varphi(u) \implies \varphi(v) \geq \varphi(\bar{u}) \implies \varphi(v) \geq \varphi(\bar{u}) + \langle 0, v-\bar{u}\rangle$

- if $\partial\varphi(u) \neq \emptyset$, then $\varphi(u) = \varphi^{**}(u)$: let $y \in \partial\varphi(u)$. Then $\varphi(v) \geq \varphi(u) + \langle y, v-u\rangle =$ convex closed function below φ. We know φ^{**} si the greatest convex closed function below φ, hence $\varphi(v) \geq \varphi^{**}(v) \geq \varphi(u) + \langle y, v-u\rangle$. If $u = v$, then we have equality.

- $y \in \partial\varphi(u) \iff \varphi(u) + \varphi^*(u) = \langle y, u\rangle$. In this case $y \in \mathrm{dom}\,(\varphi^*)$.

$'' \Longrightarrow ''$ $y \in \partial\varphi(u)$

$$\begin{aligned}
&\varphi(v) \geq \varphi(u) + \langle y, v-u\rangle \\
&\langle y, u\rangle - \varphi(u) \geq \langle y, v\rangle - \varphi(v) \\
&\langle y, u\rangle - \varphi(u) \geq \sup_v \langle y, v\rangle - \varphi(v) \\
&\langle y, u\rangle - \varphi(u) \geq \varphi^*(y)
\end{aligned}$$

Therefore, by Fenchel's inequality, we get the claim.

$'' \Longleftarrow ''$ reverse the last argument.

- if $\varphi(u) = \varphi^{**}(u)$ then $\partial\varphi(u) = \partial\varphi^{**}(u)$: if $y \in \partial\varphi^{**}(u)$, then $\varphi^{**}(v) \geq \varphi^{**}(u) + \langle y, v-u\rangle \implies \varphi^{**}(v) \geq \varphi(u) + \langle y, v-u\rangle$. We know $\varphi(v) \geq \varphi^{**}(v)$ foe every v, hence $y \in \partial\varphi(u)$, therefore $\partial\varphi^{**}(u) \subseteq \partial\varphi(u)$. If instead $y \in \partial\varphi(u)$, then $\varphi(u) < \infty$ and $\varphi(v) \geq \varphi(u) + \langle y, v-u\rangle = \varphi^{**}/u + \langle y, v-u\rangle$, which the last is a closed convex function below φ, but we know that φ^{**} is the biggest of such a function, therefore $\varphi^{**}(v) \geq \varphi^{**}(u) + \langle y, v-u\rangle$, hence $y \in \partial\varphi^{**}(u)$.

- $y \in \partial\varphi(u) \iff u \in \partial\varphi^*(u)$ and $\varphi(u) = \varphi^{**}(u)$: let $y \in \partial\varphi(u)$, by the above point, $\varphi(u) + \varphi^*(u) = \langle y, u\rangle \implies \varphi(u) \geq \varphi^{**}(u) = \sup_z \langle z, u\rangle - \varphi^*(u) \geq \langle y, u\rangle - \varphi^*(y) = \varphi(u) \implies \varphi(u) = \varphi^{**}(u)$. Recall that $\varphi^*(z) = \sup_v z^\top v - \varphi(v) \geq z^\top u - \varphi(u) = z^\top u - y^\top u + \varphi^*(y)$, hence $\varphi^*(z) \geq \varphi^*(y) + \langle u, z-y\rangle$, therefore $u \in \partial\varphi^*(y)$.

Corollary 3. *If $\varphi = \varphi^{**}$, then $y \in \partial\varphi(u) \iff u \in \partial\varphi^*(y)$ for every y.*

Chapter 18

Lecture 18 of 29/04/2026

If $\varphi(0) = \varphi^{**}(0)$ then $\partial\varphi(0) = \{y : y \text{ is a minimizer of } \varphi^*\}$.

Comment: Let φ convex and continous on $\text{int}\,(\text{dom}\,(\varphi))$. Previously we said φ restricted to $\text{ri}\,(\text{dom}\,(f))$ is continous. Consider $\varphi : \mathbb{R}^2 \to \overline{\mathbb{R}}$,

$$\varphi(x,y) = \begin{cases} 0 & \text{if } 0 \le x \le 1,\ y = 0 \\ \\ +\infty & \text{otherwise} \end{cases}$$

note that $\text{dom}\,(f) = \{(x,y) : x \in [0,1],\ y = 0\}$, so a segment, therefore $\text{int}\,(\text{dom}\,(f)) = \emptyset$. But $\text{ri}\,(\text{dom}\,(f)) = \{(x,y) : x \in (0,1),\ y = 0\} =: A$. Then $\varphi : A \to \overline{\mathbb{R}}$ is continous.

18.1 Primal/dual setting

Let $f : X \to \overline{\mathbb{R}}$, $F = X \times U \to \mathbb{R}$ such that $f(x) = F(x,0)$, $\varphi : U \to \overline{\mathbb{R}}$, $\varphi(u) = \inf_x F(x,u)$, $L : X \times Y \to \overline{\mathbb{R}}$, $g : Y \to \mathbb{R}$, $G : Y \times V \to \overline{\mathbb{R}}$ such that $g(y) = G(y,0)$, $\gamma : V \to \mathbb{R}$, $\gamma(v) = \sup_y G(y,v)$.

Denote as

$$\begin{aligned} &(P) \text{ minimize } f \text{ over } X,\ \inf_x f(x) = \inf(P) \\ &(D) \text{ maximize } g \text{ over } Y,\ \sup_y g(y) = \sup(D) \end{aligned}$$

Note that $\bar{x}$ solves $(P) \iff 0 \in \partial f(\bar{x})$ and $\bar{y}$ solves (D) if $0 \in \partial g(\bar{y})$, which the last one is a notational horror as, if g is concave, $\partial g(y)$ denotes the *superdifferential* of g at y, i.e. $z \in \partial g(y)$ if $g(w) \le g(y) + \langle z, w - y\rangle$.

We now look at L: we assert that a point $(v,u) \in \partial L(\bar{x},\bar{y})$ if $v \in \partial_x L(\bar{x},\bar{y})$ and $u \in \partial_y L(\bar{x},\bar{y})$, which means that

$$\begin{aligned} L(x,\bar{y}) &\ge L(\bar{x},\bar{y}) + \langle v, x - \bar{x}\rangle \quad \forall x \\ L(\bar{x},y) &\le L(\bar{x},\bar{y}) + \langle u, y - \bar{y}\rangle \forall y. \end{aligned}$$

Notation: we dub the relation $(0,0) \in \partial L(\bar{x}, \bar{y})$ as the *Khun-Tucker condition* for (P).

Theorem 1 (Book 15). *The implications $(a) \iff (b) \implies (c) \implies (d)$ hold among the following conditions:*

(a) $\inf(P) = \sup(D)$;

(b) $\varphi(0) = \mathrm{cl}\,(\mathrm{conv}\,(\varphi))\,(0)$;

(c) *the saddle value of L exists;*

(d) $\gamma(0) = \mathrm{cl}\,(\gamma)\,(0)$.

Assuming $F(x,\cdot)$ closed convex for all x, then $(a) \iff (b) \iff (c)$. Assuming F jointly convex, they are all equivalent. Furthermore, it holds also $(e) \implies (f)$ where

(e) *$\bar{x}$ solves (P), $\bar{y}$ solves (D),* $\inf(P) = \sup(D)$

(f) *$(\bar{x}, \bar{y})$ satisfies the Khun-Tucker condition.*

Assuming $F(x,\cdot)$ closed convex for every x, then $(e) \iff (f)$.

Proof. Not required. ∎

Corollary 2. *Suppose* $\inf(P) = \sup(D)$ *and* $\sup(D)$ *is attained. then $\bar{x}$ solves $(P) \iff \exists\, \bar{y}$ such that $(\bar{x}, \bar{y})$ satisfies the Khun-Tucker condition.*

Theorem 3. *Let $\bar{y} \in Y$, then the following are equivalent:*

(a) *$\bar{y}$ solves (D) and* $\inf(P) = \sup(D)$;

(b) $-\bar{y} \in \partial\varphi(0)$;

(c) $\inf_x L(x, \bar{y}) = \inf_x f(x)$.

Proof. We know that $-g(y) = \varphi^*(-y)$ and $\sup(D) = \varphi^{**}(0)$. Condition (a) is equivalent to $0 \in \partial\varphi^*(-y)$ because $0 \in \partial g(\bar{y})$, hence $\varphi^{**}(0) = \varphi(0)$, then apply the first proposition of today, so it follows $(a) \iff (b)$.

To prove $(a) \iff (c)$ observe that $g(\bar{y}) = \inf(P)$ is a rewrite of (c) and use the inequality $g(y) \le \inf(P)$. ∎

Theorem 4. *If F is jointly convex, then φ is convex. Assume φ is bounded on a neighbourhood of* 0, *then* $\inf(P) = \sup(D)$ *and there exists at leas one $\bar{y}$ such that $\bar{y}$ solves (D).*

Proof. Under our assumptions, φ is continous at 0, and everything follows. ■

Remark 5. We are happy also if $0 \in \operatorname{int}(\operatorname{dom}(\varphi))$.

Example 6. Considet the problem

$$\inf_x f_0(x) \quad \text{s.t } f_i(x) \leq 0.$$

Then

$$\varphi(u) = \inf_x \{f_0(x) : x \in C,\ f_i(x) \leq u_i\}.$$

Observe that $\operatorname{dom}(\varphi) = \{u \in \mathbb{R}^m : \exists x \in C \text{ s.t. } f_i(x) \leq u_i\}$. Therefore $0 \in \operatorname{dom}(\varphi)$ if there exists x such that $f_i(x) < 0$ for every i.

We will now prove strong duality for the Fenchel's problem. In this setting we have $\varphi(u) = \inf_x h(x) + K(Ax + u)$, and the F that we defined is indeed jointly convex. To have $0 \in \operatorname{dom}(\varphi)$ i can ask that there exists $\bar{x} \in \operatorname{dom}(h)$ such that K is bounded form above in a neighbourhood of $A\bar{x}$.

What about subdifferentials? $(0,0) \in \partial L(\bar{x}, \bar{y})$ if $\bar{x}$ solves $\inf(P)$ and $\bar{y}$ solves $\sup(D)$. The confition is then $L(x, \bar{y}) \geq L(\bar{x}, \bar{y})$ for any x. Recall that $L(x,y) = h(x) - K^*(-y) - y^\top Ax$. Therefore,

$$\begin{aligned} h(x) - K^*(-y) - y^\top Ax &\geq h(\bar{x}) - K^*(-y) - y^\top Ax \\ h(x) &\geq h(\bar{x}) + \langle A^\top \bar{y}, x - \bar{x}\rangle \quad \forall x. \end{aligned}$$

This condition is equivalent to $A^\top y \in \partial h(\bar{x})$. With a similar argument for the second case we get $A\bar{x} \in \partial K^*(-\bar{y})$.

Chapter 19

Lecture 19 of 05/05/2026

Example 1 (Quadratic programming)**.** Consider the problem

$$\inf_x \frac{1}{2}x^\top Qx + c^\top x \text{ s.t. } Ax \le b$$

where $Q \in \mathbb{R}^{n\times n}$ SPD, $c \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$.
The perturbation function is

$$F(x,u) = \begin{cases} \frac{1}{2}x^\top Qx - c^\top x & \text{if } Ax - b \le u \\ \\ +\infty & \text{otherwise} \end{cases}$$

The Lagrangian turns out to be

$$\begin{aligned} L(x,y) &= \inf_u F(x,u) + y^\top u \\ &= \inf_{u\in\mathbb{R}^m,\, Ax-b\le u} \frac{1}{2}x^\top Qx - c^\top x + y^\top u. \end{aligned}$$

Observe that if there exists an index k such that $y_k \le 0$ then we can pick u to be $u = (M,\dots,M, \underbrace{\lambda}_{\text{position } k}, M,\dots,M)$ and take $\lambda \to -\infty$, so $y_k\lambda \searrow -\infty$. Hence we want $y \ge 0$. Then we have to take the smallest possible entries for u that are given by $Ax - b$. Finally,

$$L(x,y) = \begin{cases} \frac{1}{2}x^\top Qx - c^\top x + y^\top(Ax-b) \text{if } y \ge 0 \\ \\ -\infty \quad \text{otherwise.} \end{cases}$$

Now we derive the dual function g:

$$\begin{aligned} g(y) &= \inf_x L(x,y) \\ &= \inf_{x\in\mathbb{R}^n} \frac{1}{2}x^\top Qx - c^\top x + y^\top(Ax-b) \text{ assuming } y \ge 0 \end{aligned}$$

Consider

$$\frac{\partial L}{\partial x}(x,y) = \nabla_x L(x,y) = Qx - c + A^\top y$$

therefore

$$\nabla_x L(x,y) = 0 \implies Qx + c + A^\top y = 0 \implies x = Q^{-1}(-c - A^\top y).$$

Notice that we are allowed to invert Q as it is SPD, therefore the inverse exists. We now substitute in L:

$$L(Q^{-1}(-c - A^\top y), y) = -\frac{1}{2}\left(c + A^\top y\right)^\top Q^{-1}\left(c + A^\top y\right) - b^\top y$$

assuming $y \geq 0$. Finally, the dual problem is

$$\sup_{y \geq 0} -\frac{1}{2}\left(c + A^\top y\right)^\top Q^{-1}\left(c + A^\top y\right) - b^\top y.$$

Do we have strong duality? Yes, observe that F is jointly convex, and it is bounded in a neighbourhood of $u = 0$.

Example 2 (Second order cone programming)**.** Consider the problem

$$\inf_{x \in \mathbb{R}^n} f^\top x$$

under the constraints $\|Ax + b\| \leq c\top x + d$, where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, $d \in \mathbb{R}$.

The perturbation function is

$$F(x,(u,z)) = \begin{cases} f^\top x & \text{if } \|Ax + b - z\| - c^\top x - d \leq u \\ \\ +\infty & \text{otherwise} \end{cases}$$

with $z \in \mathbb{R}^m$ and $u \in \mathbb{R}$.

The Lagrangian is then

$$L(x,(y,w)) = \inf_{(u,z): \|Ax - b - z\| - c^\top x - d \leq u} f^\top x + yu + w^\top z$$

where we call (y, w) the *"dual variable"*, $y \in \mathbb{R}$, $w \in \mathbb{R}^m$. Again, if $y < 0$ then we get $-\infty$. We take then u as small as possible, so $u = \|Ax + b - z\| - c^\top x - d$. Therefore

$$
\begin{aligned}
L(x,(y,w)) &= \inf_z f^\top x + y\left(\|Ax+b-z\| - c^\top - d\right) + w^\top z \\
&= f^\top x - yc^\top x - yd + \inf_z w^\top z + y\|Ax+b-z\| \quad \tilde{z} = Ax+b-z \\
&= f^\top x - yc^\top x - yd + \inf_{\tilde{z}} w^\top\left(Ax+b-\tilde{z}\right) + y\|\tilde{z}\| \\
&= f^\top x - yc^\top x - yd + w^\top\left(Ax+b\right) + \inf_{\tilde{z}} -w^\top \tilde{z} + y\|\tilde{z}\| \\
&= f^\top x - yc^\top x - yd + w^\top\left(Ax+b\right) - \sup_{\tilde{z}} w^\top \tilde{z} - y\|\tilde{z}\| \\
&= f^\top x - yc^\top x - yd + w^\top\left(Ax+b\right) - y\left(\sup_z \left(\frac{w^\top}{y}\right) z - \|z\|\right).
\end{aligned}
$$

In parenthesis we have the conjugate of the norm function $\left(\|\cdot\|\right)^*\left(\frac{w}{y}\right)$. We recall that the convex conjugate of the norm function is the convex indicator function of the unit ball. Indeed $\left(\|\cdot\|\right)(v) = \sup_{z\in\mathbb{R}^n} v^\top z - \|z\|$. Recall that, in ℓ^2 it holds that

$$
\|x\| = \sup_{\|y\|\le 1} x^\top y.
$$

Therefore,

$$
\begin{aligned}
\left(\|\cdot\|\right)^*(v) &= \sup_{z\in\mathbb{R}^n} v^\top z - \|z\| \\
&= \sup_{\|z\|=1,\,\lambda\ge 0} \lambda v^\top z - \lambda \\
&= \sup_{\lambda\ge 0} \lambda\left(\sup_{\|z\|=1} v^\top z - 1\right) \\
&= \sup_{\lambda\ge 0} \lambda\left(\|v\| - 1\right) \\
&= \begin{cases} 0 & \text{if } \|v\| \le 1 \\ \\ +\infty & \text{otherwise} \end{cases} \qquad = \delta_{\{(y,w):\|w/y\|\le 1\}}.
\end{aligned}
$$

In the end,

$$
L(x,(y,w)) = f^\top x - yc^\top x - yd + w^\top(Ax+b) + = \delta_{\{(y,w):\|w/y\|\le 1\}}.
$$

The dual function is then

$$
\begin{aligned}
g((y,w)) &= \inf_x L(x,(y,w)) \\
&= \inf_x \left(f - yc + A^\top w\right)^\top - yd + w^\top b + = \delta_{\{(y,w):\|w/y\|\le 1\}} \\
&= \begin{cases} -yd + w^\top b = \delta_{\{(y,w):\|w/y\|\le 1\}} & \text{if } f - yc + A^\top w = 0 \\ \\ -\infty \quad \text{otherwise.} \end{cases}
\end{aligned}
$$

The dual problem is then

$$\sup_{(y,w)} -yd + w^\top b$$

under the constraints $f - yc + A^\top w = 0$, $\|w\| \le |y|$.

Chapter 20

Lecture 20 of 06/05/2026

20.1 Proximal mappings

The topic is now the *proximal mappings*, we will consider the book Convex Analysis by Dirk Lorentz and we will see proximal algorhitms and proximal mappings (chapter 8 + piece 10 + piece 12).

Definition 1 (Proximal mapping/Monreau envelope)**.** Let $f : \mathbb{R}^d \to \overline{\mathbb{R}}$ proper convex closed, $\lambda \geq 0$. We define the *proximal mapping* the map $\text{prox}_{(\lambda f)} : \mathbb{R}^d \to \mathbb{R}^d$ defined as

$$\text{prox}_{(\lambda f)}(x) = \text{argmin}_{y \in \mathbb{R}^d} \lambda f(y) + \frac{1}{2}\|y - x\|^2.$$

We can think it as a "minimizer of the rescaled function λf + quadratic term". We define the *Monreau envelope* as

$$M_{\lambda f}(x) = \inf_{y \in \mathbb{R}^d} f(y) + \frac{1}{2\lambda}\|y - x\|^2.$$

Theorem 2. *Let $f : \mathbb{R}^d \to \overline{\mathbb{R}}$ be proper convex closed and fix $\lambda > 0$. The following holds:*

1. *for every $x \in \mathbb{R}^d$ there exists a unique $\hat{x}$ such that $\hat{x} = \text{prox}_{(\lambda f)}(x)$;*
2. *$\hat{x}$ as above is characterized by the variational inequality*
$$\langle x - \hat{x}, y - \hat{x} \rangle + \lambda f(\hat{x}) - f(y) \leq 0$$
for every $y \in \mathbb{R}^d$;
3. *x^* is a minimizer of f if and noly if x^* is a fixed point of $\text{prox}_{(\lambda f)}$, i.e. $x^* = \text{prox}_{(\lambda f)}(x^*)$;*
4. *the Monreau envelope $M_{\lambda f}$ is a convex differentiable function and $\nabla M_{\lambda f} = \frac{1}{\lambda}\Big(x - \text{prox}_{(\lambda f)}(x)\Big)$;*

5. *it holds that*

$$\operatorname{argmin}_x f(x) = \operatorname{argmin}_x M_{\lambda f}(x).$$

Proof. It's a bluff. ∎

Example 3. Let C be nonempt convex closed, $f(x) = \delta_C(x)$ the convex indicator function. Then

$$\begin{aligned}\operatorname{prox}_{(\lambda f)}(x) &= \operatorname{argmin}_{y \in \mathbb{R}^d} \lambda \delta_C(y) + \frac{1}{2}\|y - x\|^2 \\ &= \operatorname{argmin}_{y \in C} \frac{1}{2}\|y - x\|^2 = P_C(x)\end{aligned}$$

the projection function of C. Moreover,

$$\begin{aligned}M_{\lambda f} &= \frac{1}{2\lambda}\|P_C(x) - x\|^2 \\ &= \frac{1}{2\lambda} d(x, C)^2\end{aligned}$$

where $d(x, C) = \inf_{y \in C}\|x - y\|$.

Example 4. Let $f(x) = |x|$, so $\hat{x} = \operatorname{prox}_{(\lambda f)}(x) = \operatorname{argmin}_y \lambda\,|y| + \frac{1}{2}(y - x)^2$.

Observe that if $x > 0$ then $\hat{x} > 0$ and $\operatorname{prox}_{(\lambda f)}(-x) = -\operatorname{prox}_{([)}\lambda f](x)$, so we can compute only for $x \geq 0$, with $y \geq 0$. The minimum of the objective function is achived when its derivative is 0 (notice that the "inside function" is differentiable), hence $\lambda + y - x = 0 \implies y = x - \lambda$.

All in all,

$$\operatorname{prox}_{(\lambda|\cdot|)}(x) = \begin{cases} x - \lambda & \text{if } x \geq \lambda \\ 0 & \text{if } -\lambda < x < \lambda \\ x + \lambda & \text{if } x \leq -\lambda \end{cases} = \max(|x| - \lambda, 0)\operatorname{sgn}(x).$$

The Monreau envelope is the

$$M_{\lambda f}(x) = \begin{cases} |x| - \frac{\lambda}{2} & \text{if } |x| \geq \lambda \\ \\ \frac{1}{2\lambda}x^2 & \text{if } |x| < \lambda \end{cases}$$

which is called *Huber function*, a differentiable appproximation of $|\cdot|$.

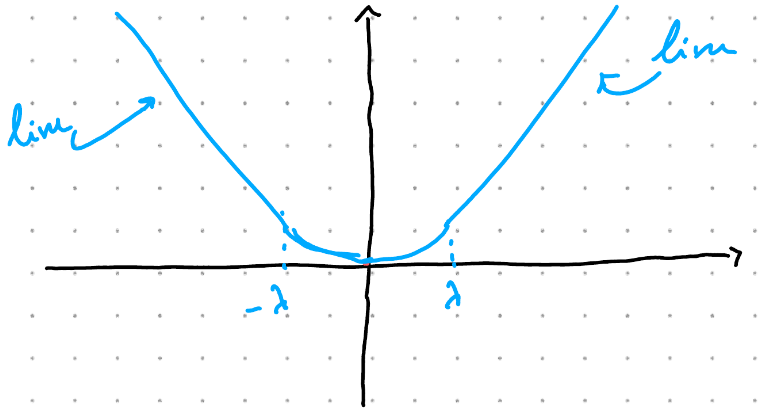


Figure 20.1: Monreau envelope of $|\cdot|$ function

PROPOSITION 5. 1. *If* $g(x) = \tau f(\mu x)$, $\tau, \mu \in \mathbb{R}$, $\tau > 0$ *then*

$$\text{prox}_{(\lambda g)}(x) = \frac{1}{\mu}\text{prox}_{(\lambda\mu^2\tau f)}(x);$$

2. *if* $g(x) = f(Qx)$ *with* Q *orthonormal matrix, then*

$$\text{prox}_{(\lambda g)}(x) = Q^\top \text{prox}_{(\lambda f)}(Qx).$$

20.1.1 Monreau identity

THEOREM 6 (Monreau identity)**.** *Let* $f : \mathbb{R}^d \to \overline{\mathbb{R}}$ *proper closed convex,* $f^*(y) = \sup_x y^\top x - f(x)$*. Then*

$$x = \text{prox}_{(f)}(x) + \text{prox}_{(f^*)}(x).$$

To prove this we need the following:

LEMMA 7. *Let* $f, g : \mathbb{R}^d \to \overline{\mathbb{R}}$ *proper closed convex. Then if there exists* $\bar{x} \in \text{dom}\,(f) \cap \text{dom}\,(g)$ *with* f *continous at* $\bar{x}$ *we have* $\partial(f+g)(x) = \partial f(x) + \partial g(x)$.

Proof. The proof is not difficoult but very technical and not short, uses Hahn-Banach, no further details. ■

Proof. (Monreau identity)
Fix $\hat{x} = \text{prox}_{(f)}(x)$, then

$$
\begin{aligned}
& 0 \in \partial \left(f(\cdot) + \frac{1}{2}\|\cdot - x\|^2 \right)(\hat{x}) \\
\implies & 0 \in \partial f(\hat{x}) + \frac{1}{2}(\underbrace{\hat{x} - x}_{=\nabla\|\cdot\|^2}) \\
\implies & x - \hat{x} \in \partial f(\hat{x}).
\end{aligned}
$$

Recall that since f is proper convex, then $f = f^{**}$. Let $y = x - \hat{x}$. Then $y \in \partial f(\hat{x}) \iff \hat{x} \in \partial f^*(y)$, hence

$$
\begin{aligned}
& x - y \in \partial f^*(y) \\
\implies & 0 \in \partial f^*(y) + (y - x) \\
\implies & 0 \in \partial \left(f^*(\cdot) + \frac{1}{2}\|\cdot - x\|^2 \right)(y)
\end{aligned}
$$

hence $y = \operatorname{prox}_{(f^*)}(x)$. In the end,

$$
\operatorname{prox}_{(f)}(x) + \operatorname{prox}_{(f^*)}(x) = \hat{x} + y = \hat{x} + x - \hat{x} = x.
$$

■

In general $x = \operatorname{prox}_{(\lambda f)}(x) + \operatorname{prox}_{\left(\frac{1}{\lambda} f^*\right)}\left(\frac{x}{\lambda}\right)$.

Chapter 21

Lecture 21 of 12/05/2026

21.1 Proximal algorhitms

We know that x^* minimizes $f \iff x^*$ is a fixed point of $\text{prox}_{(\lambda f)}$.

Idea: x^0 given, iterate a fixed point iteration $x^{k+1} = \text{prox}_{(\lambda f)}(x^k)$. We know that this works if $\text{prox}_{(\lambda f)}$ is a contraction!

Proposition 1. *The proximal mapping is 1-Lipschits continous (assuming f convex proper closed, $\lambda > 0$), i.e.* $\|\text{prox}_{(\lambda f)}(x) - \text{prox}_{(\lambda f)}(y)\| \leq \|x - y\|$.

Proof.

$$\hat{x} = \text{prox}_{(\lambda f)}(x) \implies \langle x - \hat{x}, \hat{y} - \hat{x} \langle + \lambda \left(f(\hat{x}) - f(\hat{y})\right) \leq 0$$
$$\hat{y} = \text{prox}_{(\lambda f)}(y) \implies \langle y - \hat{y}, \hat{x} - \hat{y} \rangle + \lambda \left(f(\hat{y}) - f(\hat{x})\right) \leq 0.$$

Taking the sum of the two, we end up with the relation

$$\langle \hat{x} - x, \hat{x} - \hat{y} \rangle + \langle y - \hat{y}, \hat{x} - \hat{y} \rangle \leq 0.$$

Consider $\|\hat{x} - \hat{y}\|^2$, then

$$\begin{aligned} \|\hat{x} - \hat{y}\|^2 &= \langle \hat{x} - \hat{y}, \hat{x} - \hat{y} \quad \pm x, \pm y \text{ in the first term and expand} \\ &= \langle \hat{x} - x, \hat{x} - \hat{y} \rangle + \langle y - \hat{y}, \hat{x} - \hat{y} \rangle + \langle x - y, \hat{x} - \hat{y} \rangle \\ &\leq \langle x - y, \hat{x} - \hat{y} \rangle \leq \|x - y\| \cdot \|\hat{x} - \hat{y}\|. \end{aligned}$$

Therefore $\|\hat{x} - \hat{y}\| \leq \|x - y\|$.

■

21.1.1 Proximal point method

x^0 given, $x^{k+1} = \text{prox}_{(t_k f)}(x^k)$ for $\{t_k\}_k$ a given sequence of steps. Recall that $\nabla M_{\lambda f}(x) = \frac{1}{\lambda}(x - \text{prox}_{(\lambda f)}(x)) \implies x^{k+1} = x^k - t_k \nabla M_{t_k f}(x^k)$.

Proposition 2. $f(x^{k+1}) \leq f(x^k) - \frac{1}{t_k}\|x^{k+1} - x^k\|^2$.

Proof. Immidiate consequence of variational inequality. ∎

Proposition 3. *Let f be proper closed convex, $f_{\mathrm{opt}} = f(x^*) = \min_x f(x)$. Then it holds that*

$$\begin{aligned} \|x^{k+1} - x^*\| &\leq \|x^k - x^*\| \\ f(x^{N+1}) - f_{\mathrm{opt}} &\leq \frac{\|x^0 - x^*\|}{2\sum_{k=1}^N t_k} \end{aligned}$$

where N is the total number of steps.

21.2 Splitting methods

Assume that we can decompose the objective function as $f(x) + g(x)$, where f is "simple in the prox sense", i.e. it is easy to compute $\mathrm{prox}_{(\lambda f)}$ and g is smooth.

Our main assumption is $g : \mathbb{R}^d \to \mathbb{R}$ convex and differentiable. Moreover, we assume ∇g to be L-Lipschits continous, $L > 0$, i.e.

$$\|\nabla g(x) - \nabla g(y)\| \leq L\|x - y\|.$$

In this case we say that g is L-smooth.

Lemma 1. *If $g : \mathbb{R}^d \to \mathbb{R}$ is L-smooth then*

$$g(y) - g(x) - \langle \nabla g(x), y - x\rangle \leq \frac{L}{2}\|x - y\|^2.$$

Proof.

$$\begin{aligned} g(y) - g(x) - \langle \nabla g(x), y - x\rangle &= \int_0^1 \langle \nabla g(x + \tau(y-x)) - \nabla g(x), y - x\rangle \,\mathrm{d}\tau \\ &= \int_0^1 \|\nabla g(x + \tau(y-x)) - \nabla g(x)\| \cdot \|y - x\| \,\mathrm{d}\tau \\ &\leq L \int_0^1 \|\tau(y-x)\| \cdot \|y - x\| \,\mathrm{d}\tau \\ &\leq \frac{L}{2}\|y - x\|^2 \end{aligned}$$

∎

Let now $\lambda > 0$, define $x^+ = x - \lambda \nabla g(x)$ a descend step for g. By the above lemma,

$$
\begin{aligned}
g(x^+) &\leq g(x) + \langle \nabla g(x), x^+ - x \rangle + \frac{L}{2}\|x^+ - x\|^2 && (21.1)\\
&\leq g(x) + \langle \nabla g(x), -\lambda \nabla g(x) \rangle + \frac{L}{2}\|\lambda \nabla g(x)\|^2 && (21.2)\\
&= g(x) - \lambda \|\nabla g(x)\|^2 + \lambda^2 \frac{L}{2}\|\nabla g(x)\|^2 && (21.3)\\
&= g(x) - \lambda \left(1 - \frac{L}{2}\lambda\right) \|\nabla g(x)\|^2 && (21.4)
\end{aligned}
$$

which means that $g(x^+) \leq g(x)$ provided that $0 \leq \lambda \leq 2/L$. What about $f + g$? Substitute g with an upper bound,

$$
g(x^k) + \langle \nabla g(x^k), x - x^k \rangle + \frac{1}{2\lambda}\|x - x^k\|^2
$$

in the spirit of the above lemma.
Then

$$
\begin{aligned}
x^{k+1} &= \mathrm{argmin}_x \left[f(x) + g(x^k) + \langle \nabla g(x^k), x - x^k \rangle - \frac{1}{2\lambda}\|x - x^k\|^2 \right]\\
&= \mathrm{argmin}_x \left[f(x) + \langle \nabla g(x^k), x - x^k \rangle - \frac{1}{2\lambda}\|x - x^k\|^2 \right].
\end{aligned}
$$

Observe that

$$
\begin{aligned}
\langle \lambda \nabla g(x^k), x - x^k \rangle &= \frac{1}{2}\|x - x^k\| + 2\langle \lambda \nabla g(x^k), x - x^k \rangle + \|\lambda \nabla g(x^k)\|^2 - \|\lambda \nabla g(x^k)\|^2\\
&= \frac{1}{2}\|x - (x^k - \lambda \nabla g(x^k))\|^2 - \frac{1}{2}\lambda^2 \|\nabla g(x^k)\|.
\end{aligned}
$$

In the end, $x^{k+1} = \mathrm{argmin}_x \left[\lambda f(x) + \frac{1}{2}\|x - (x^k - \lambda \nabla g(x^k))\|^2 \right]$. Therefore,

$$
x^{k+1} = \mathrm{prox}_{(\lambda f)}(x^k - \lambda \nabla g(x^k)).
$$

This iteration is called proximal gradient method.

Example 2 (Regularized least squares)**.** We are interested in

$$
\min_{x \in \mathbb{R}^n} \|Ax - b\|^2
$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$. Define $g(x) = \frac{1}{2}\|Ax - b\|^2$, so that $\nabla g(x) = A^\top (Ax - b) \implies A^\top Ax = A^\top b$ is the minimum point.

The data b could be noisy or corrupted, so we add a regularizing term. The problem becomes

$$\min_x \frac{1}{2}\|Ax - b\|^2 + \alpha R(x)$$

where the first takes the role of $g(x)$ and the second $f(x)$.

21.3 Primal-dual Chambolle-Pock algorithm

We are interested in the problem

$$\min_x h(x) + K(Ax)$$

where h, K are convex proper closed functions, A a matrix. The idea is: instead of optimize directly, we look at the Lagrangian and we try to optimize for a saddle point.

So, given (x^k, y^k) optimize for $\max_y L(x^k, y)$. Observe that

$$\max_y h(x^k) - K^*(-y) - y^\top Ax^k = -\min_y K^*(y) - y^\top Ax^k.$$

Define $y^{k+1} \approx -\mathrm{argmin}_y K^*(y) - y^\top Ax^k$ as done via the proximal gradient step on y, therefore

$$y^{k+1} = -\mathrm{prox}_{(\sigma K^*)}(y^k + \sigma Ax^k).$$

Given then (x^k, y^{k+1}) we do a primal descend step on x, hence

$$\begin{aligned}
x^{k+1} &\approx \mathrm{argmin}_x L(x, y^{k+1}) && (21.5)\\
&\approx h(x) - \underbrace{K^*(-y^{k+1})}_{\text{not depending on } x} - (y^{k+1})^\top Ax && (21.6)\\
&\approx \mathrm{argmin}_x h(x) - (y^{k+1})^\top Ax && (21.7)\\
\implies x^{k+1} &= \mathrm{prox}_{(\tau h)}(x^k + A^\top y^{k+1}). && (21.8)
\end{aligned}$$

The final algorithm then reads: choose $\tau, \sigma > 0$, $\vartheta \in [0,1]$, (x^0, y^0) and set $\bar{x}^0 = x^0$. Then iterate

$$
\begin{aligned}
y^{k+1} &= \operatorname{prox}_{(\sigma K^*)}(y^k + \sigma A \bar{x}^k) \\
x^{k+1} &= \operatorname{prox}_{(\tau h)}(x^k - \tau A^\top y^{k+1}) \\
\bar{x}^{k+1} &= x^{k+1} + \vartheta(x^{k+1} - x^k).
\end{aligned}
$$

Chapter 22

Lecture 22 of 13/05/2026

Do we have convergence? Just a few words: if we choose $\theta = 1$, set $L = \|A\|$ ans it holds that $\tau\sigma L^2 < 1$, then $(x^k, y^k) \to (x^*, y^*)$ as C/N (order 1 convergence). The proof is quite long and not very interesting.

Example 1 (ROF model, image denoising)**.** We can consider an image (black and white) as a map $g : \Omega \to [0, 1]$. Imagine that in the acquisition process for g an arbitrary amount of noise is added. We can recover the "true" image as the minimizer of the functional

$$\frac{\lambda}{2} \int_\Omega |u - g|^2 \, \mathrm{d}x + \int_\Omega |Du|$$

where $|Du|$ denotes the total variation measure. In general we seek for a minimizer $u \in \mathrm{BV}(\Omega)$, but, if $u \in W^{1,1}(\Omega)$, then $|Du| = \|\nabla u\| \mathrm{d}x$

We consider first the function u to be a $W^{1,1}$ function, then, WLOG, $\Omega = [0, 1] \times [0, 1]$. We discretize Ω with a uniform grid in each dimension, so that $\Omega \approx (x_i, y_j)_{ij} = (ih, jh)_{ij}$ where h is the discretization step. Then we define the function on the grid as a matrix of samples $u = \left\{u_{ij}\right\}_{ij}$ and the noisy function $g = \left\{g_{ij}\right\}_{ij}$. The problem then becomes

$$\frac{\lambda}{2} \sum_{ij} h^2 \left|u_{ij} - g_{ij}\right| + \sum_{ij} h^2 \|(\nabla u)_{ij}\|.$$

Let $X = \mathbb{R}^{M \times N}$, $Y = \left(\mathbb{R}^{M \times N}, \mathbb{R}^{M \times N}\right)$, $p \in Y \implies p = (p^1, p^2)$, and define the map $A : X \to Y$, where $Au = (\nabla u)_{ij}$ with foreward finite differences, so

$$(\nabla u)^1_{ij} = \begin{cases} \dfrac{u_{i+1,j} - u_{ij}}{h} & \text{if } i < M \\ \\ 0 \quad \text{otherwise, Neumann BC} \end{cases}$$

and

$$(\nabla u)^2_{ij} = \begin{cases} \dfrac{u_{i,j+1} - u_{i,j}}{h} & \text{if } j < N \\ \\ 0 & \text{otherwise, Neumann BC} \end{cases}$$